



北京大学 计算科学成像实验室

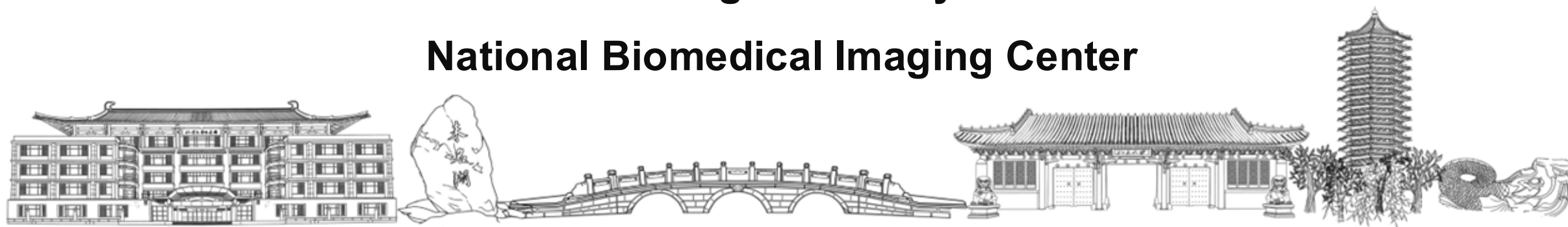
PEKING UNIVERSITY COMPUTATIONAL SCIENTIFIC IMAGING GROUP

Adaptive Optics: Principles & Applications

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National Biomedical Imaging Center



What is Adaptive Optics?

Adaptive optics (AO) is an ensemble of electro-optical and computational methods that aim to recover the optimal performance of an optical system.

— *Nature Reviews Methods Primers 2021*

Science & Art of manipulating light physically & algorithmically

It is by nature **interdisciplinary**, related to optics, signal processing, control, etc.

Computational AO was proposed more than 20 years ago for correcting aberrations by ... correcting the effects of the aberrations using deconvolution...

Physical correction is preferred over deconvolution because the additional measurement of the corrected image yields additional photons, substantially increasing the signal to noise ratio ...

This fact applies to AO in general and should not be understated.

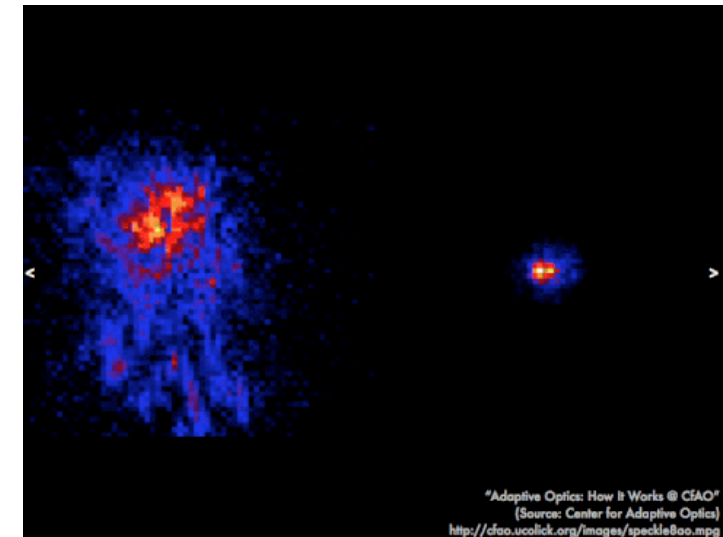
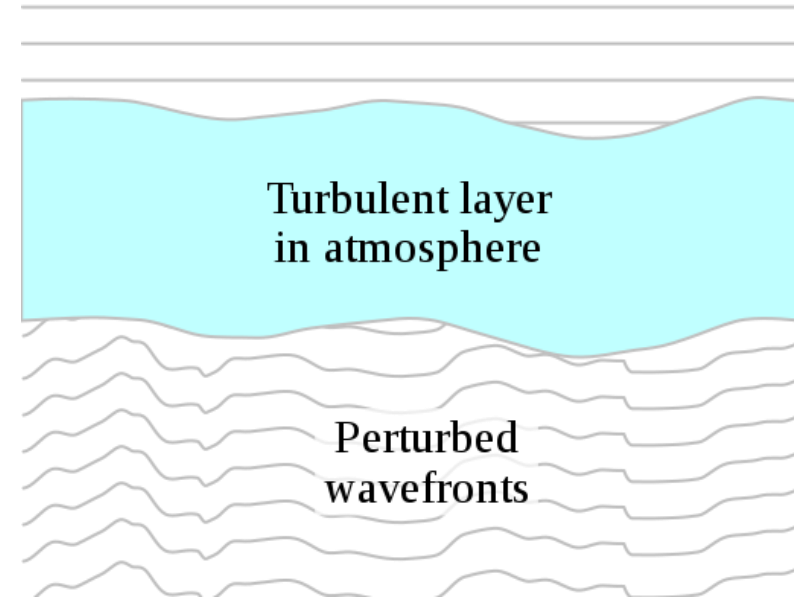
— *Nature Reviews Methods Primers* 2021

History of Adaptive Optics – Newton's Concern

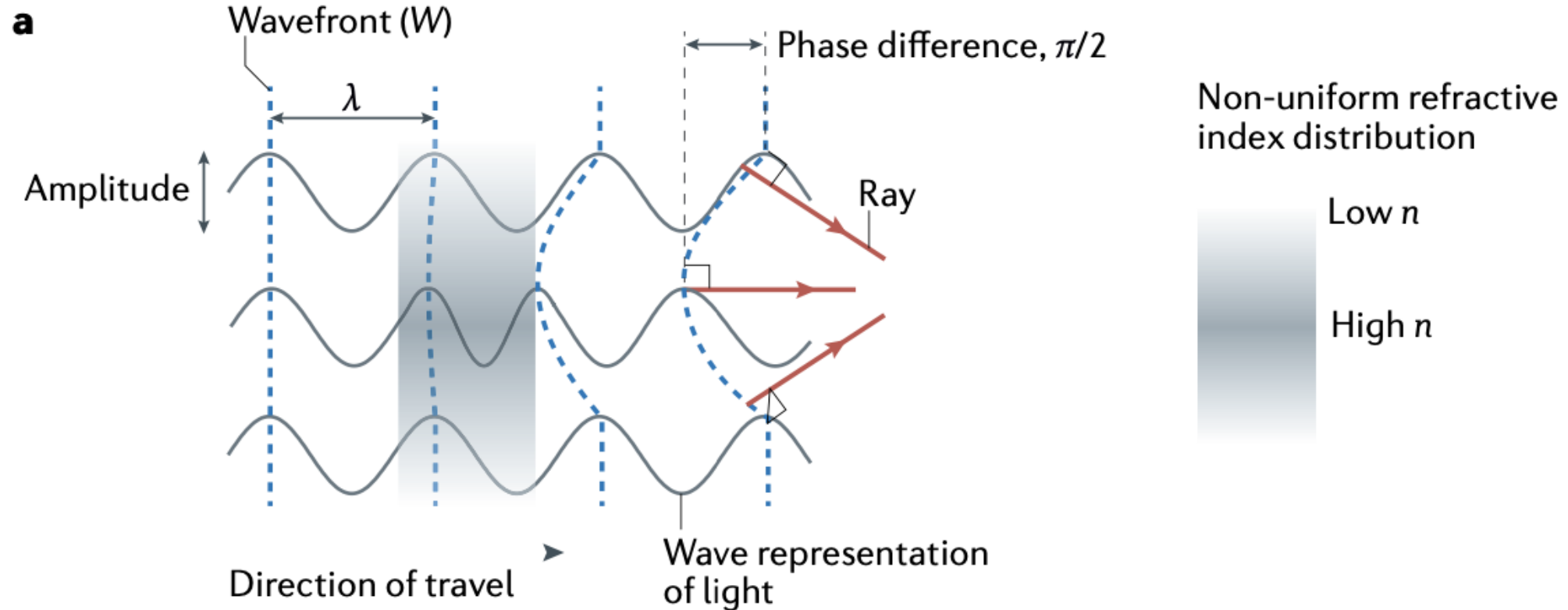
Isaac Newton, writing in *Opticks* in 1730, has realized the effects of atmospheric turbulence on image formation.

If the Theory of making Telescopes could at length be fully brought into Practice, yet there would be certain Bounds beyond which Telescopes could not perform. For the Air through which we look upon the Stars, is in perpetual Tremor; as may be seen by the tremulous Motion of Shadows cast from high Towers, and by the twinkling of the fix'd Stars. But these Stars do not twinkle when viewed through Telescopes which have large apertures. For the Rays of Light which pass through divers parts of the aperture, tremble each of them apart, and by means of their various and sometimes contrary Tremors, fall at one and the same time upon different points in the bottom of the Eye, and their trembling Motions are too quick and confused to be perceived severally. And all these illuminated Points constitute one broad lucid Point, composed of those many trembling Points confusedly and insensibly mixed with one another by very short and swift Tremors, and thereby cause the Star to appear broader than it is, and without any trembling of the whole. Long Telescopes may cause Objects to appear brighter and larger than short ones can do, but they cannot be so formed as to take away that confusion of the Rays which arises from the Tremors of the Atmosphere. The only Remedy is a most serene and quiet Air, such as may perhaps be found on the tops of the highest Mountains above the grosser Clouds.

Plane waves from distant point source



Wavefront Aberrations



Wavefront is defined by the spatial locations that share the same phase.

Optical aberrations can be understood as deformations of the light's wavefront from its perfect form - planar for a collimated beam or spherical for a focusing beam - that produces the sharpest image

Astronomical Seeing & Diffraction Limit

Seeing is the degradation of the image of an astronomical object due to turbulence in the atmosphere of Earth that may become visible as blurring, twinkling or variable distortion.

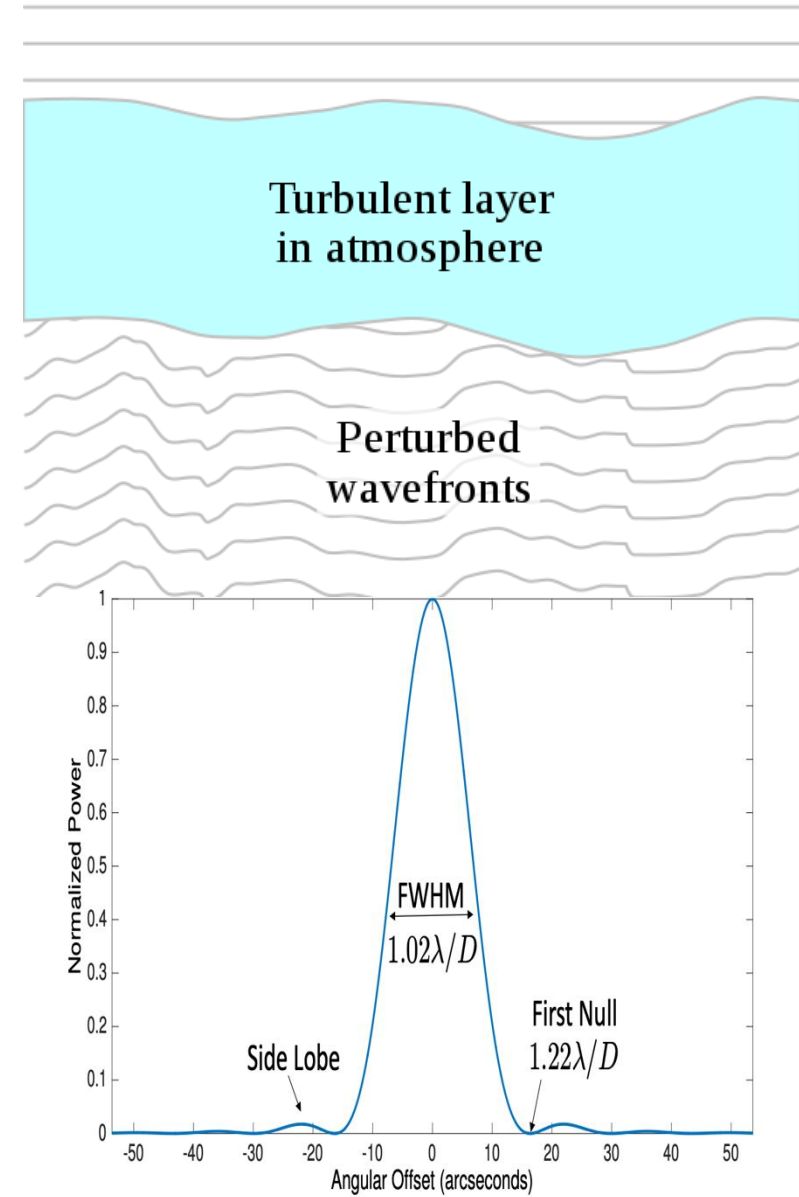
The model assumes that the wavefront perturbations are brought about by **variations in the refractive index of the atmosphere**.

Diffraction-limited image (Airy-pattern image)

$$R = 1.22 f \lambda / D$$

Seeing is a major limitation to the angular resolution in astronomical observations with telescopes that would otherwise be limited through diffraction by the size of the telescope aperture.

Plane waves from distant point source



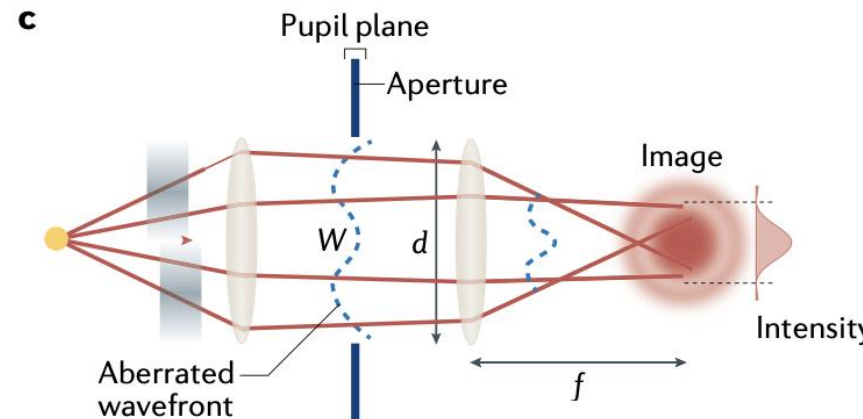
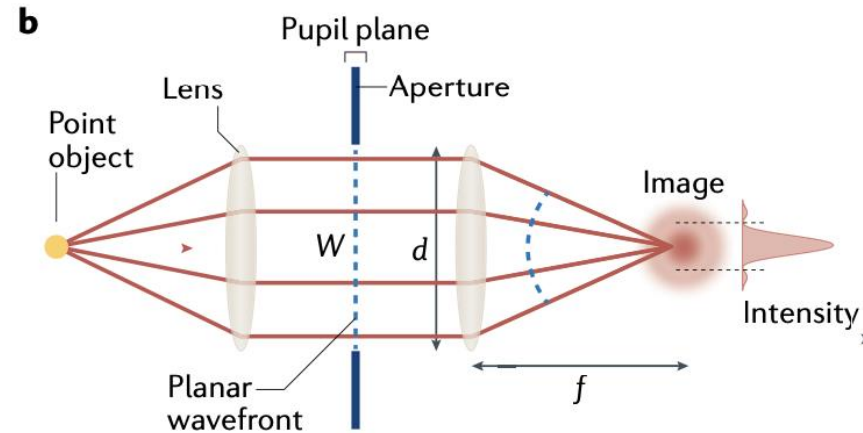
Statistical measure of atmospheric turbulence – r_0 & t_0

Coherent length or Fried parameter (r_0)

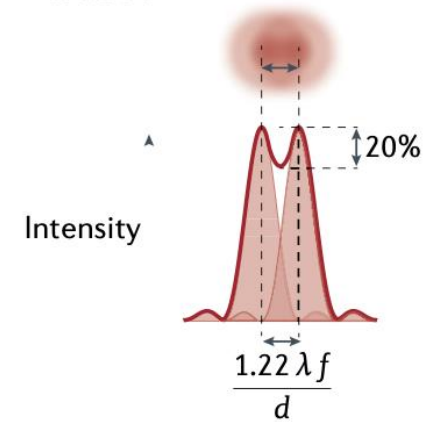
- a statistical measure of the turbulence's spatial frequency
- around **10cm** at visible wavelength at good site
- telescope diameter $< r_0$, diffraction limit defined by the telescope size;
- telescope diameter $> r_0$, diffraction limit defined by r_0

Coherent time (t_0)

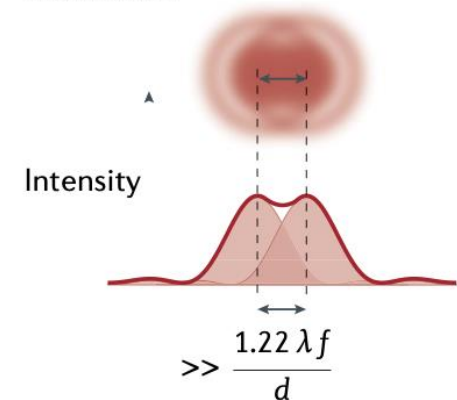
- a statistical measure of the turbulence's jiggling rate
- on the order of **1ms**



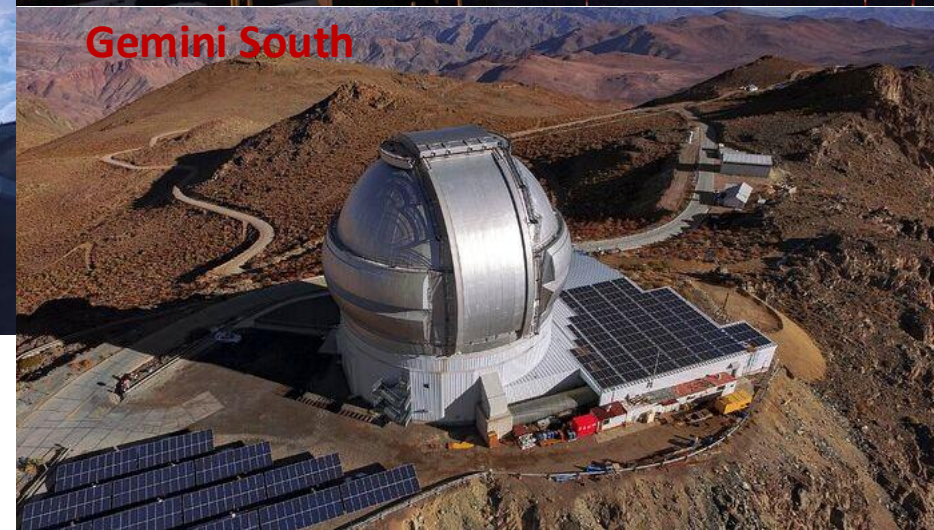
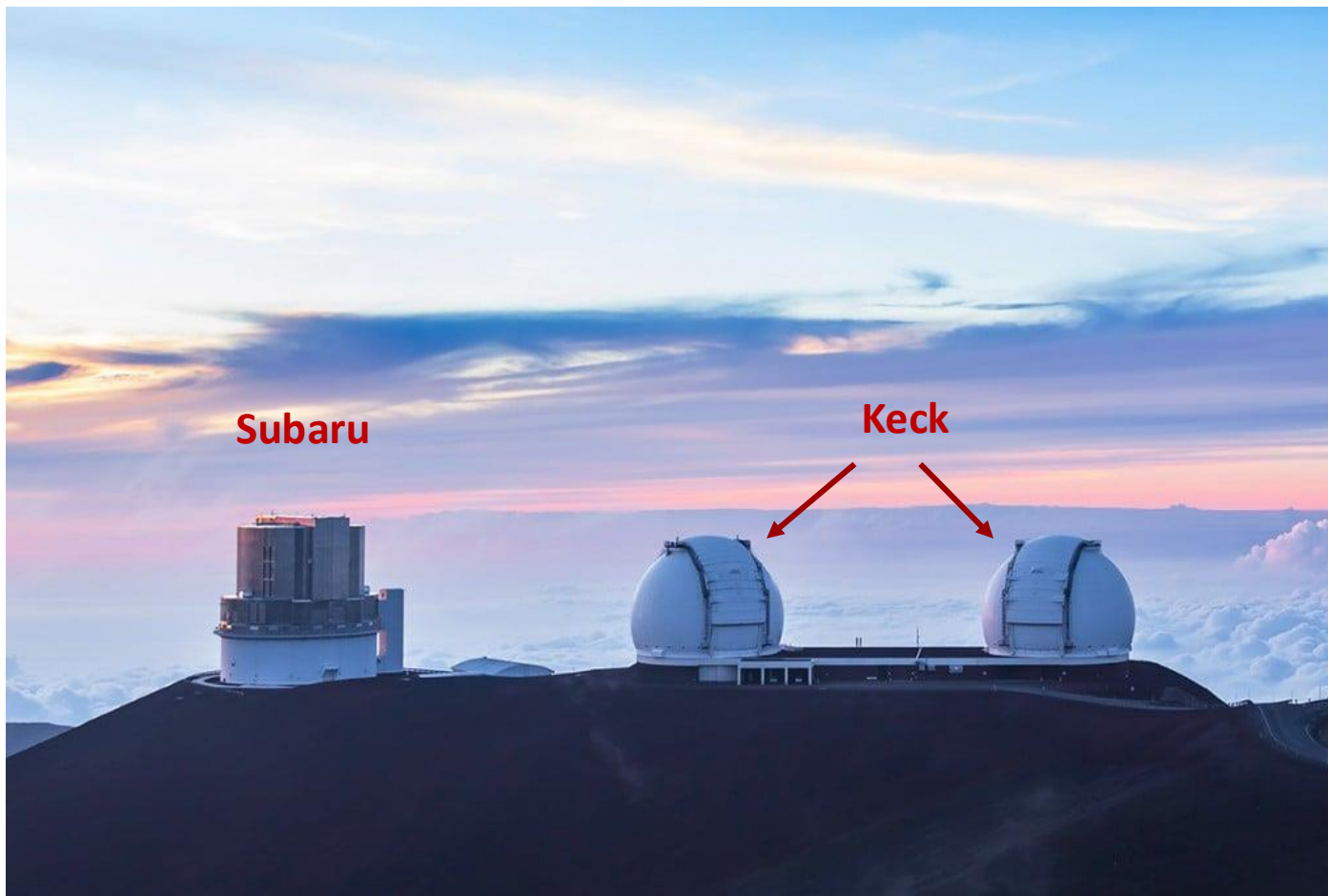
Resolution



Resolution



Newton's Proposal



Build telescopes “on the tops of highest mountains above the grosser cloud”.

- *Isaac Newton*

Newton's Proposal



Build telescopes “on the tops of highest mountains above the grosser cloud”.

- Isaac Newton



青海冷湖赛什腾山 (Lenghu)

- a world-class site on the Qinghai-Tibetan Plateau for the construction of an optical astronomy observatory.
- an altitude of 4,200-4,500 meters
- optical observation conditions similar to top-tier observatories in the Western Hemisphere

[*Deng, L., Yang, F., Chen, X. et al. Lenghu on the Tibetan Plateau as an astronomical observing site. Nature 596, 353–356 \(2021\).*](#)

1950s - Babcock's & Linnik's Proposal

- 1953 AO seminal paper: *"The Possibility of Compensating Astronomical Seeing", PASP*
- All basic building component is here
 - Beam Splitter
 - Wavefront Sensor & Corrector
 - Close-loop Arrangement

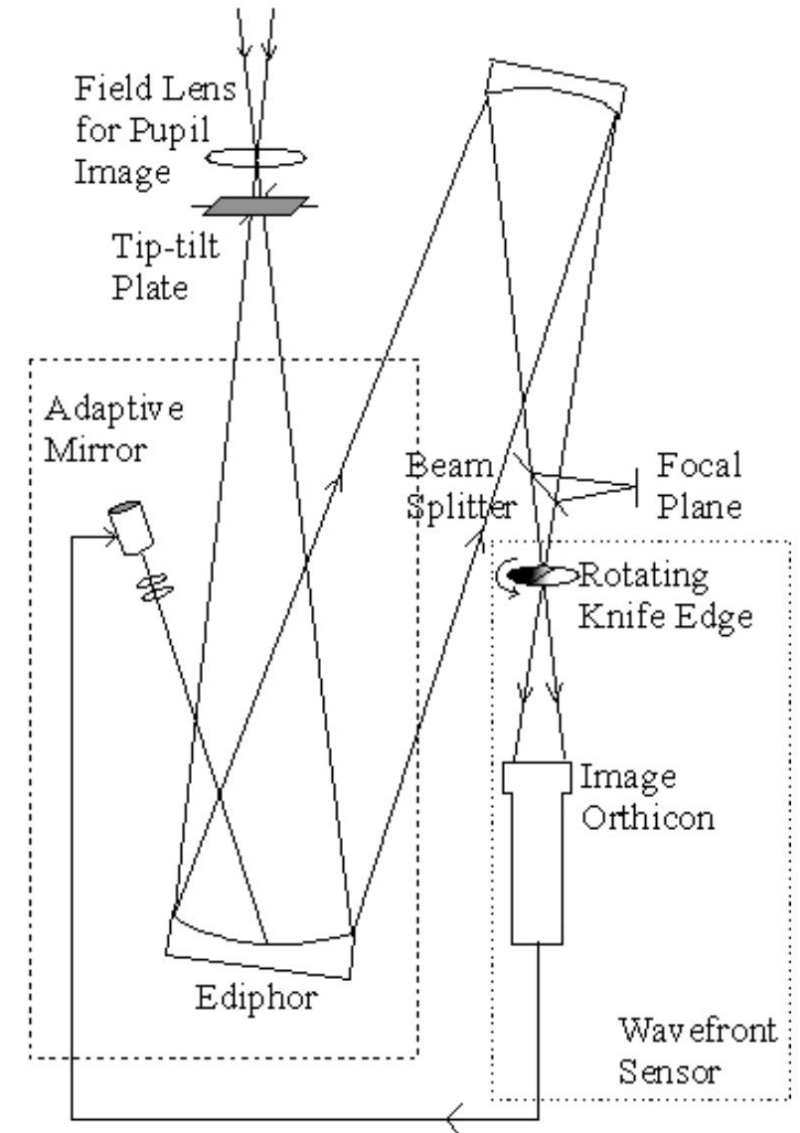


Horace Babcock (U.S.A.)



V. P. Linnik (U.S.S.R)

- 1957 Independent proposal for adaptive optics systems and their limitations, *Opt. Spectrosc.*
- First describe the artificial guide star



Babcock's Design

1960s – 1980s - AO Development During Cold War Years



AO off



AO on

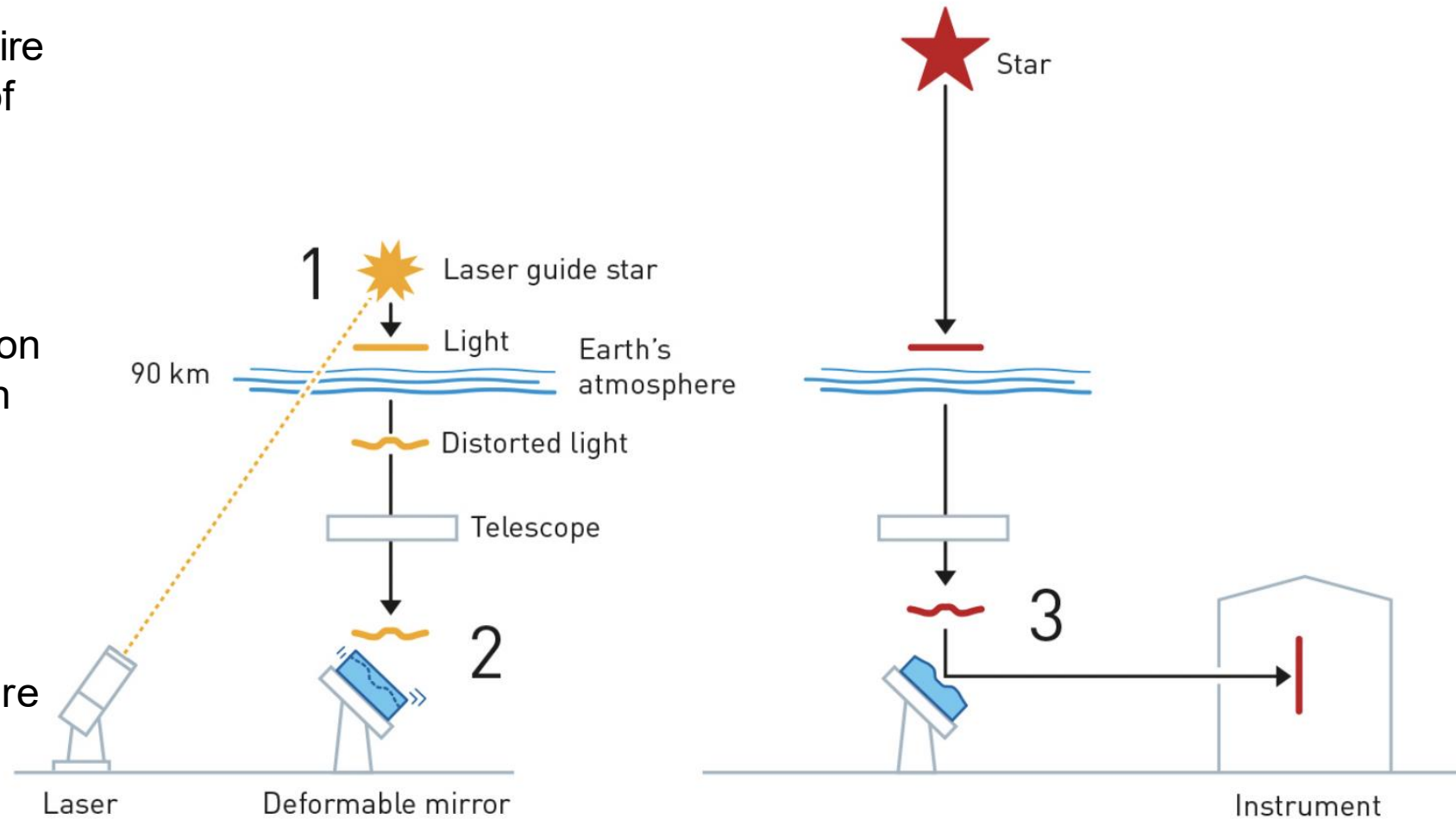


AO on +
post-processing

- 1970s: **DARPA** (Defense Advanced Research Projects Agency) looks for a mean of identifying/imaging USSR **low earth orbit satellites**.
 - Imaging fast moving target ($>10\text{kHz}$) in any seeing conditions, visible wavelength.
 - Many technologies to develop:
 - fast wave-front sensing (shearing interferometer)
 - Reconstructor (analog, array of resistor; local)
 - DM: Feinlieb et al $\rightarrow$ MPM* 21 actuators
 - More AO theory: WFSs for white light, extended sources.
 - 1973-1974: the Real-Time Atmospheric Compensator (RTAC).
 - 1973: First close loop lab test ever. Very successful.
 - 1974: First RTAC field tests. Successful.
- 1980s: Strategic Defense Initiative (SDI) – **Star War Program**
 - Ballistic Missile Defense
 - Anti-Satellite
- 1990s: Many technologies declassified due to the end of cold war

1990s - Astronomical AO

- Astronomical AO systems require a wavefront reference source of light called a **guide star**.
- **Natural guide star (NGS)**
 - Ideal point source for wavefront sensing/correction
 - Bright stars not available in all parts of sky
- **Laser guide star (LGS)**
 - Created by shining a laser into atmosphere
 - Can be positioned anywhere in the sky



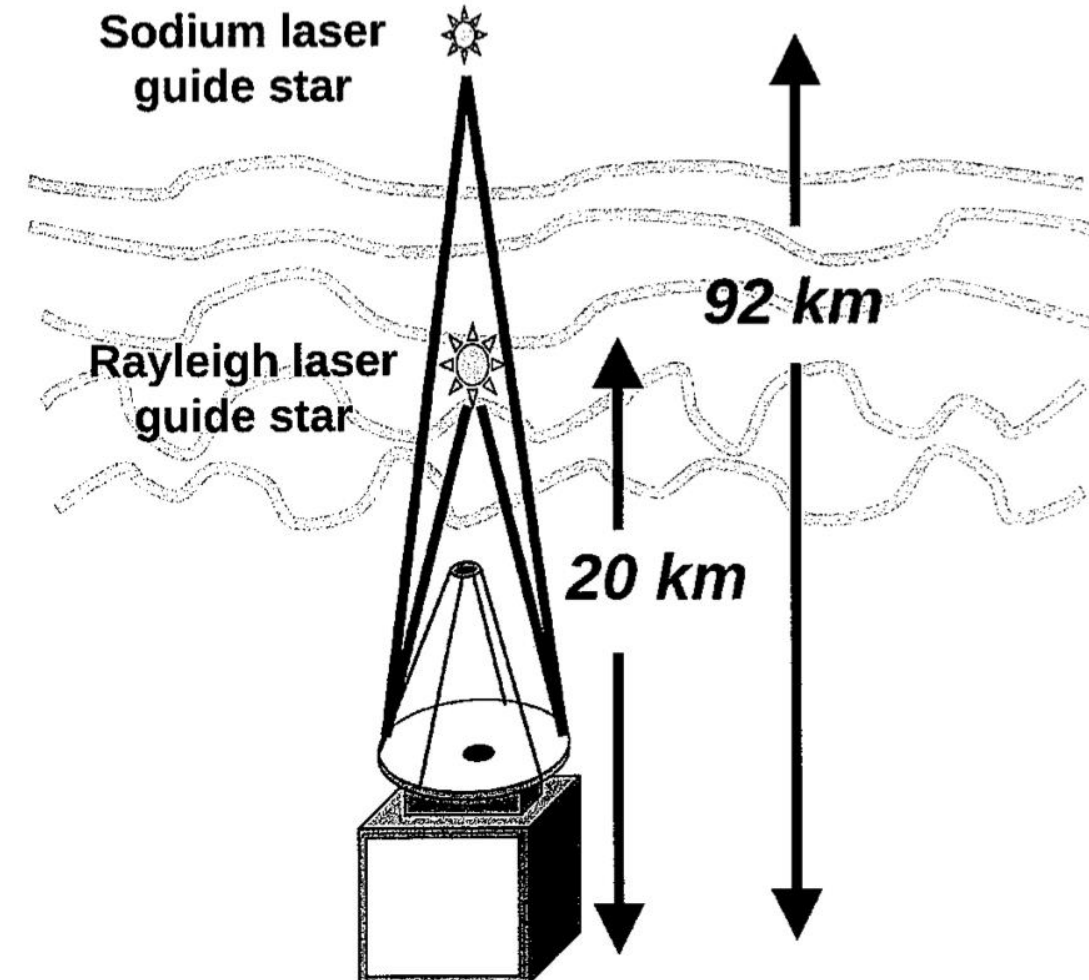
Laser Guide Star: Rayleigh Beacon v.s. Sodium Beacon

Rayleigh beacons

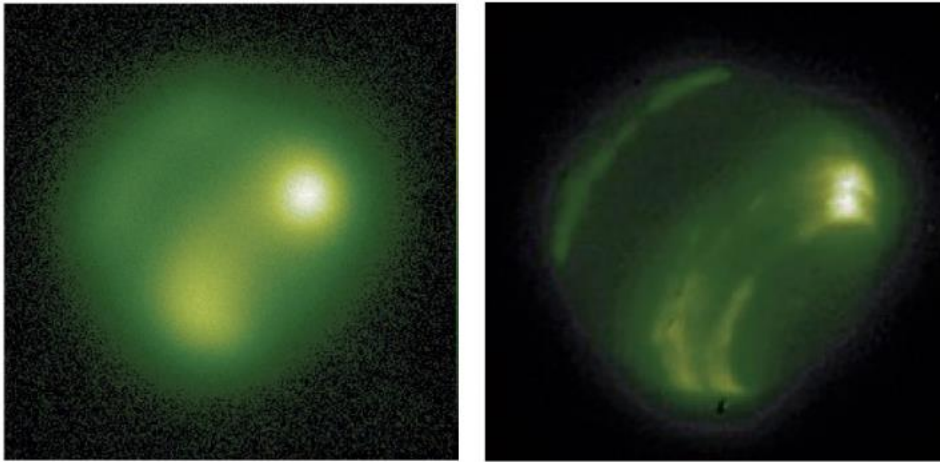
- rely on the scattering of light by the molecules in the lower atmosphere around 20km altitude.
- simple and inexpensive
- cone effect: only measures the portion of atmosphere below the guide star

Sodium beacons

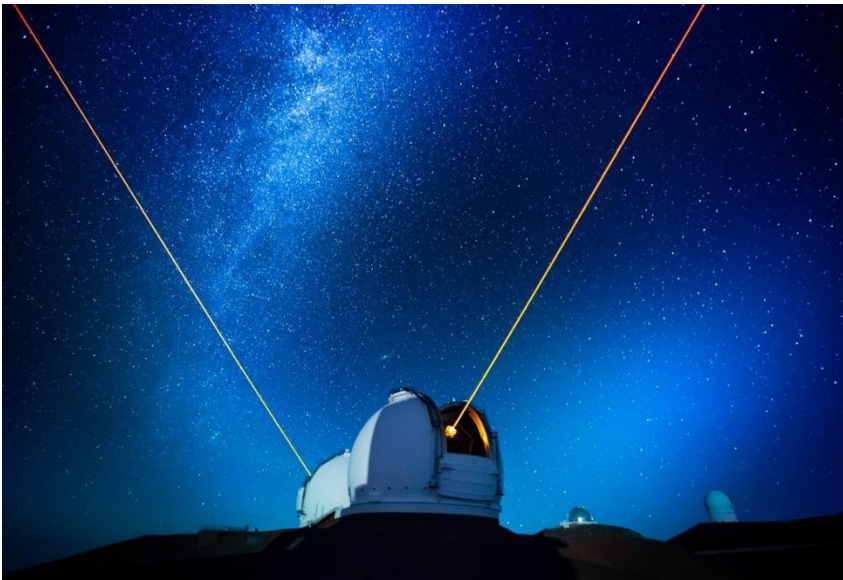
- created by using a laser tuned to 589.2 nanometers to energize atoms in the sodium layer of the mesosphere at around 90 km altitude. The sodium atoms then re-emit the laser light, producing a glowing artificial star.



Keck Telescope AO



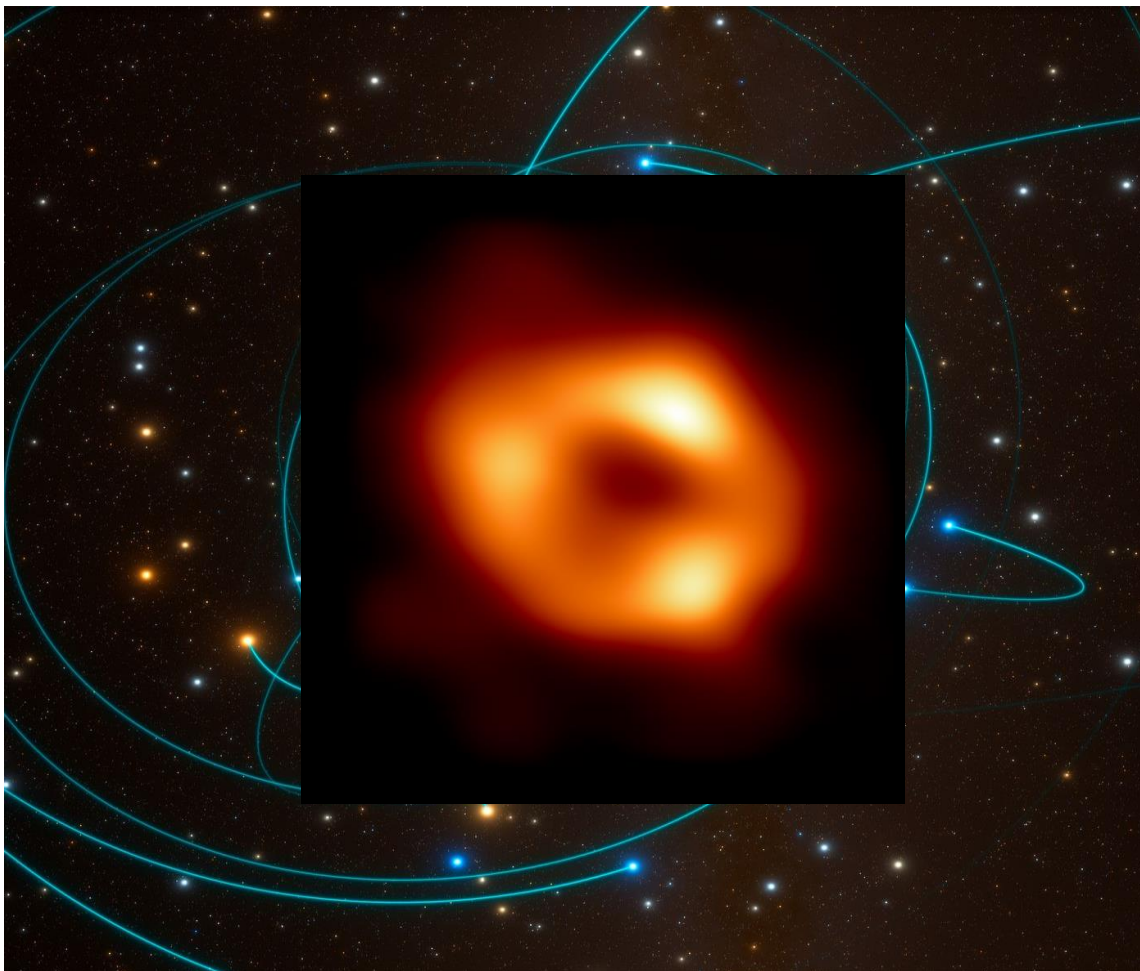
Neptune before and after AO correction



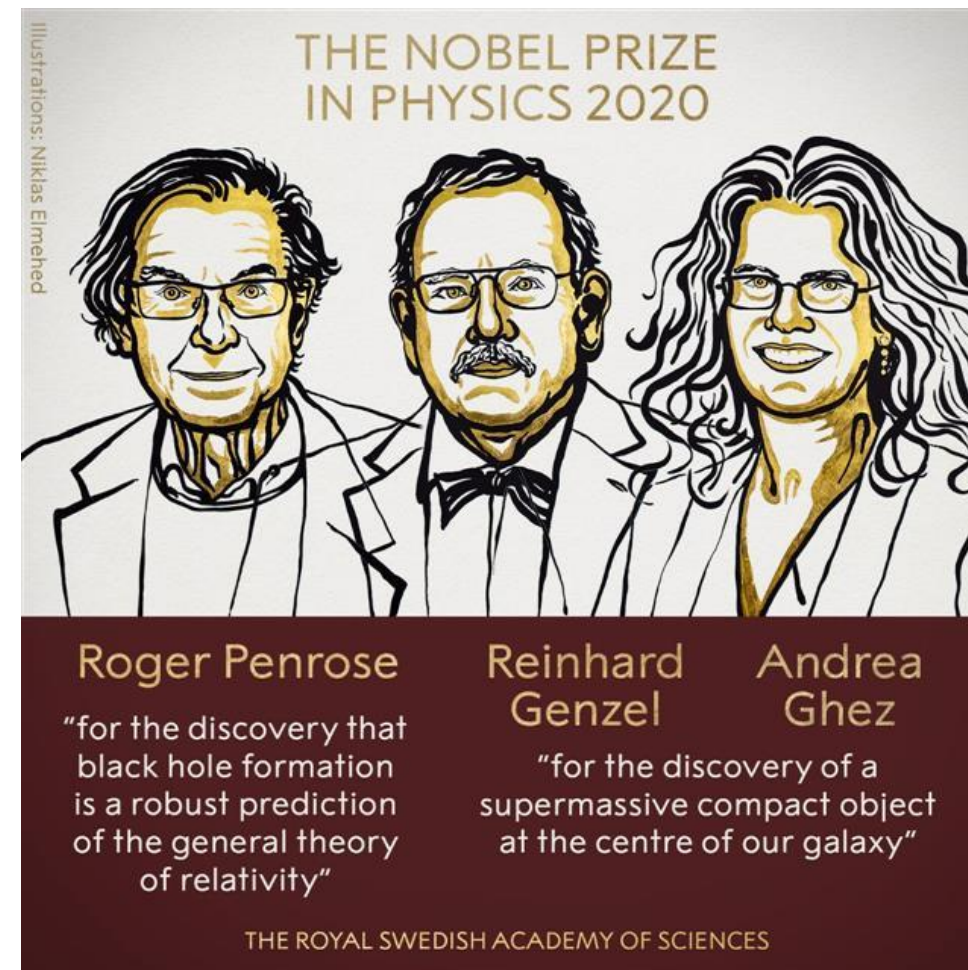
W. M. Keck Observatories

- 1999: First light for AO on the Keck II telescope.
 - First AO system for a new generation of 8–10 m telescopes,
 - natural guide star system
 - consisted of a separate tip–tilt corrector, a 349 actuator deformable mirror and a SH wavefront sensor
 - Strehl ratios of up to 0.37 in H band ($1.6\ \mu\text{m}$)
 - dramatic improvements in image quality afforded by AO on large ground-based telescopes. The original Keck AO systems
- 2001: sodium LGS installed
 - significantly improved sky coverage
- 2008: wavefront sensor and control upgraded
 - Improved the sampling, frame rate and latency
 - Improved Strehl ratio and faint star limits for natural guide star and LGS modes⁸⁶

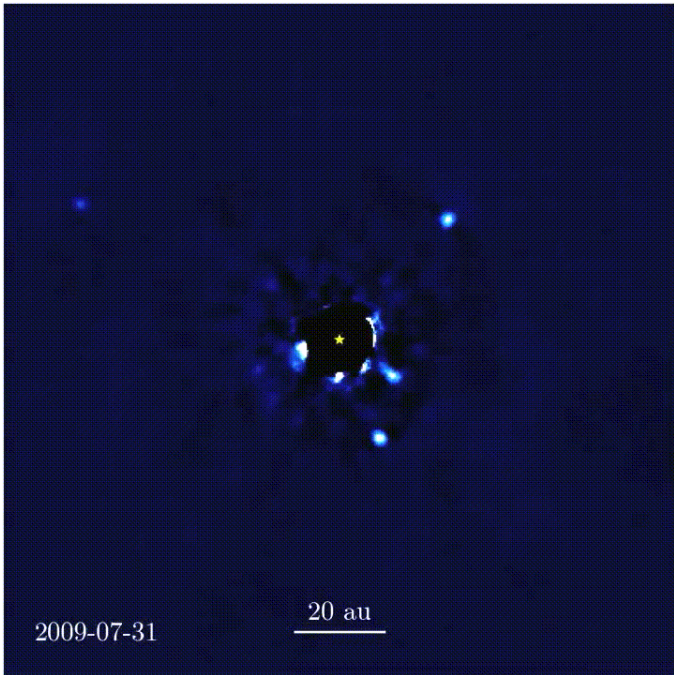
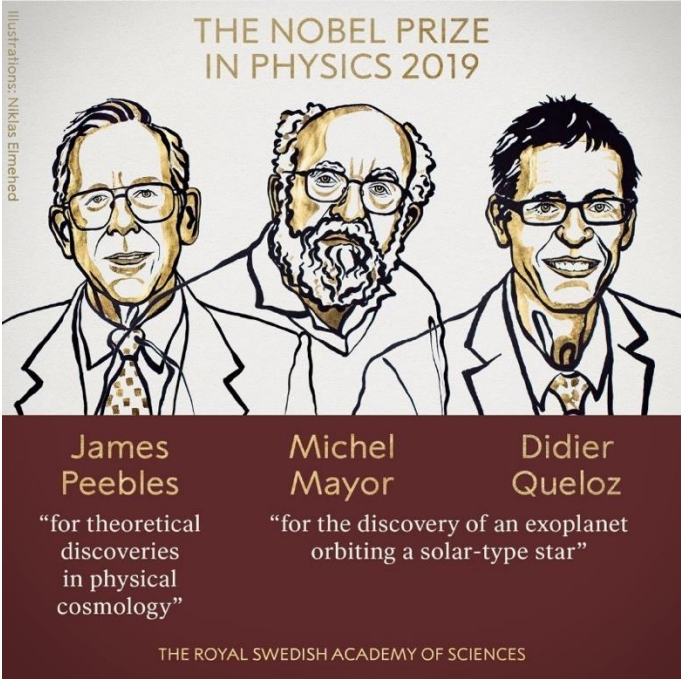
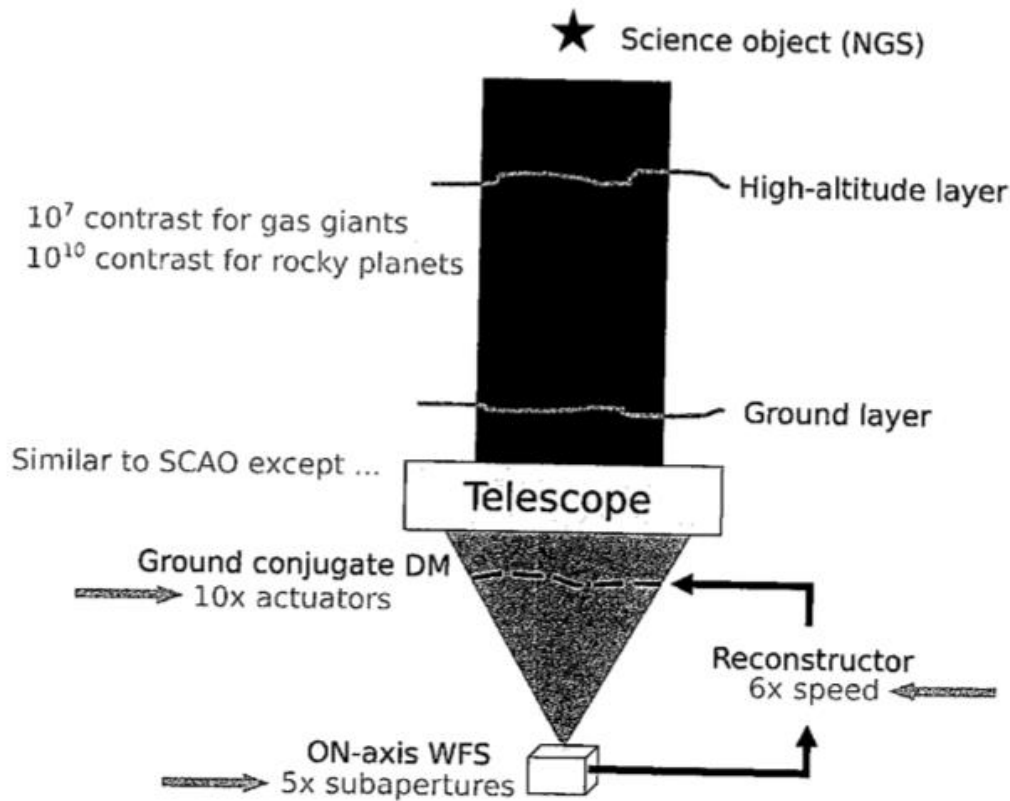
AO for Black Hole Discovery



Keck/UCLA Galactic Center Group



Extreme AO – High Contrast Imaging of Exoplanets



HR8799 bcde

The **science target** acts as the **NGS**.

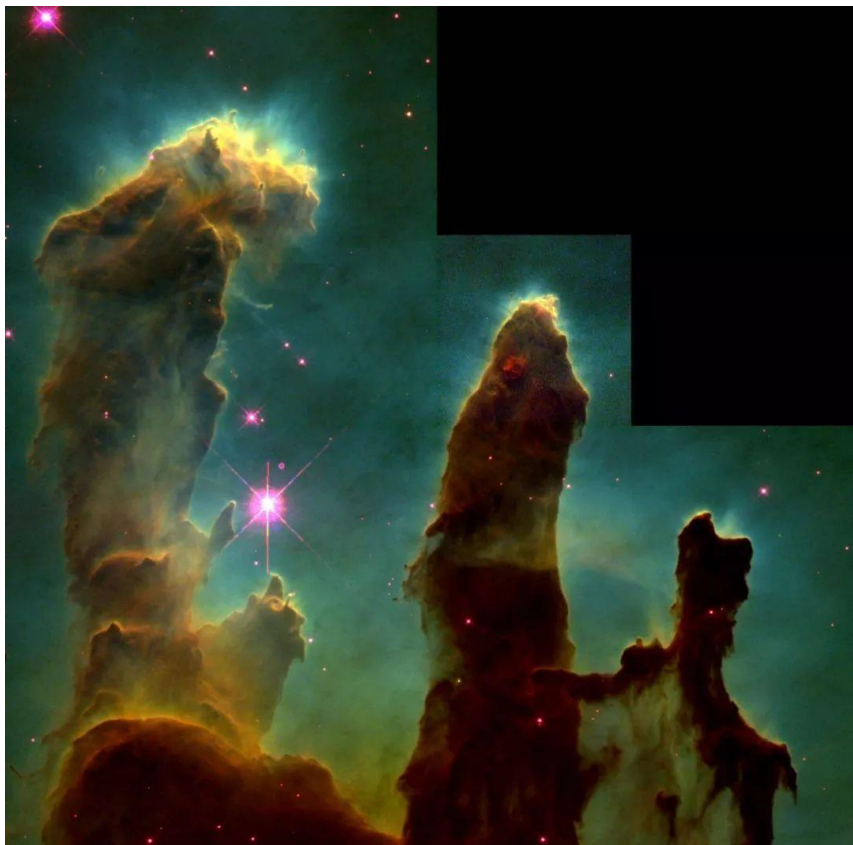
AO cannot replace space facilities

Hubble Space Telescope



- **AO pros**
 - much cheaper
- **AO cons**
 - FOV < 1arcmin
 - Atmosphere opaque at some wavelengths (UV, NIR, etc)
 - 1000-10000x brighter sky background from OH emission
- Proposed in 1946
- Funded in 1970s
- Launched on April 24, 1990
- Entered service on May 20, 1990
- the most productive astronomical telescope ever
- **Still in service**

Success of Hubble Space Telescope (HST)

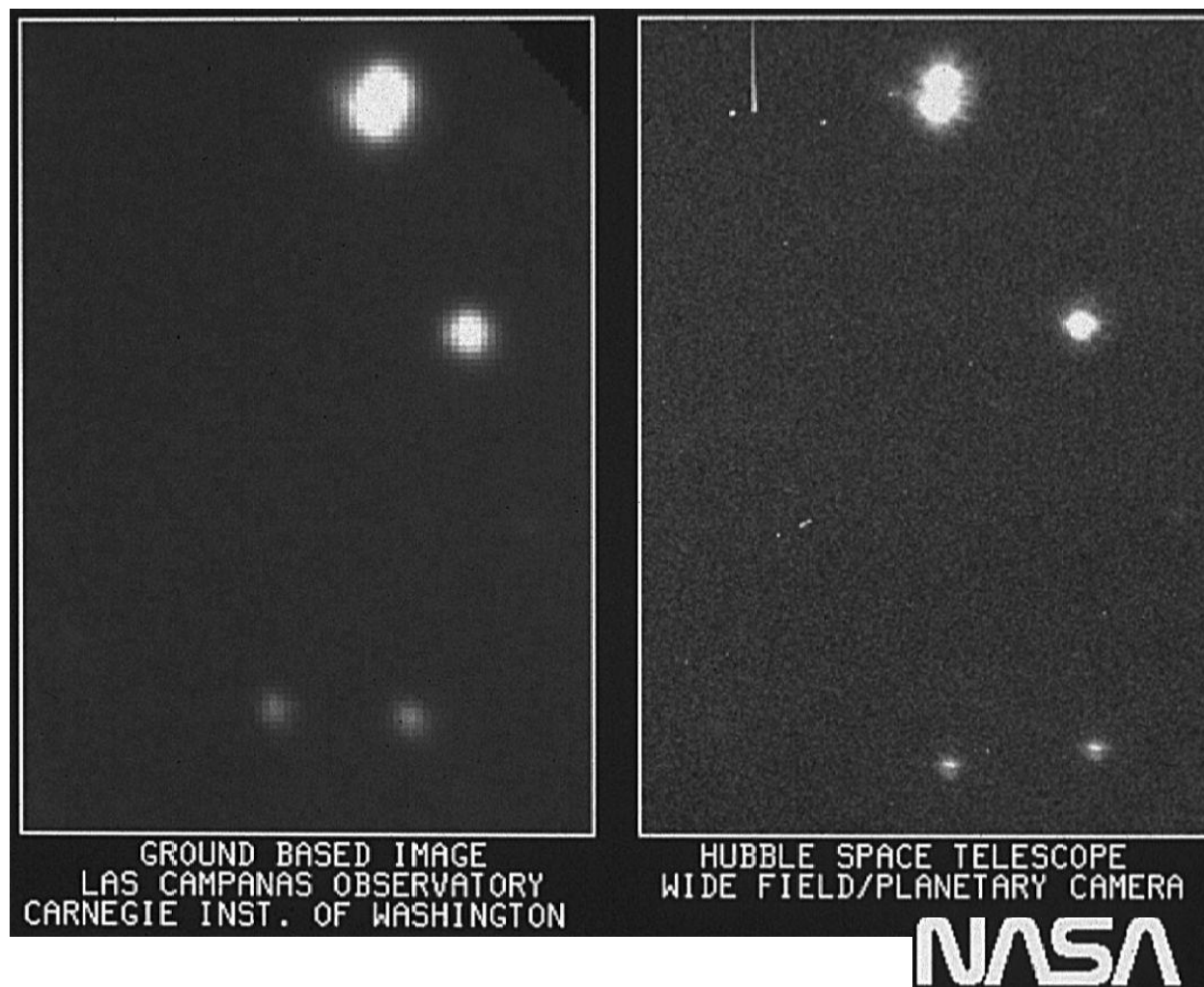


Star birth in Eagle Nebula
(Nov 2, 1995)

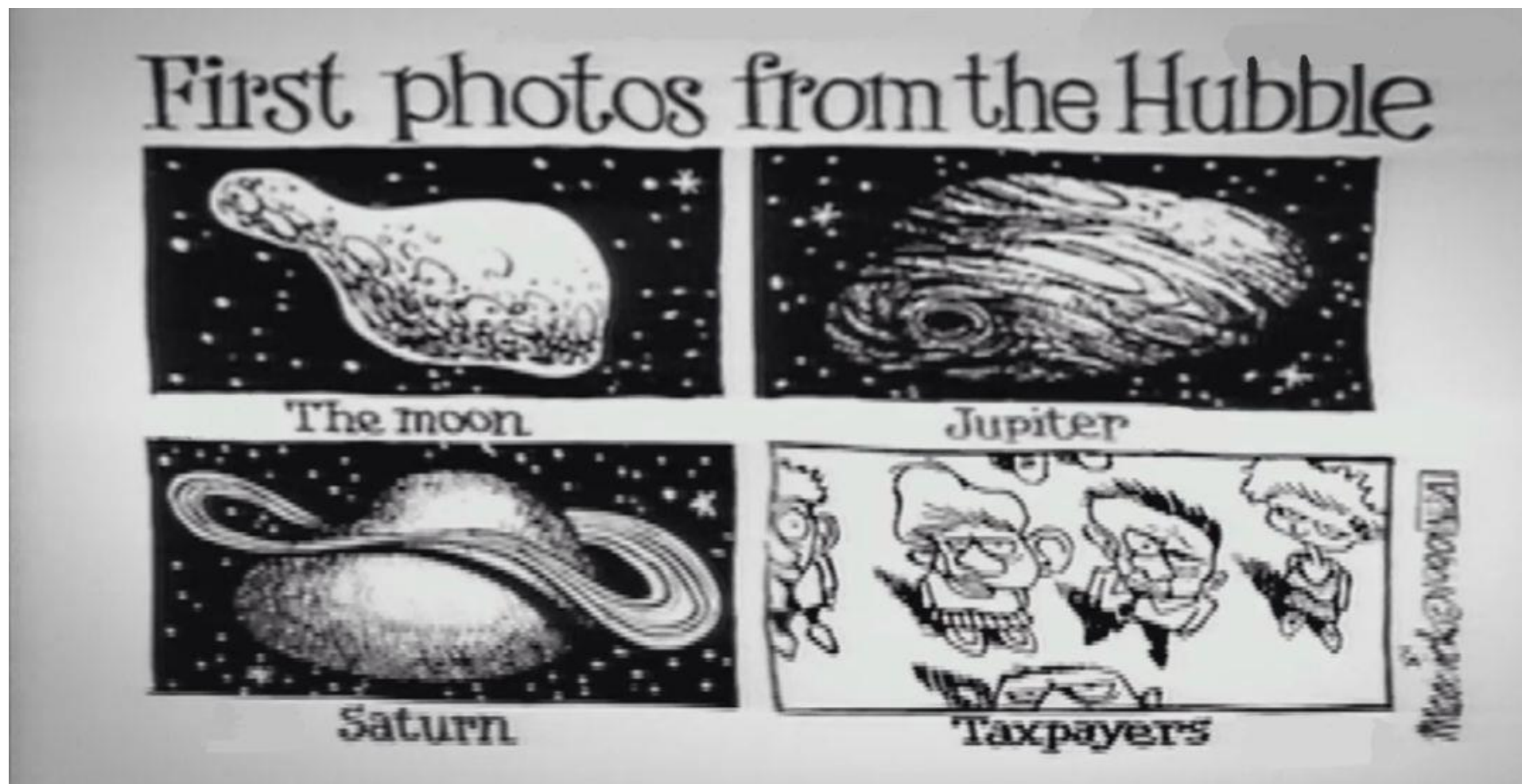


Hubble deep field image
(Jan 15, 1996)

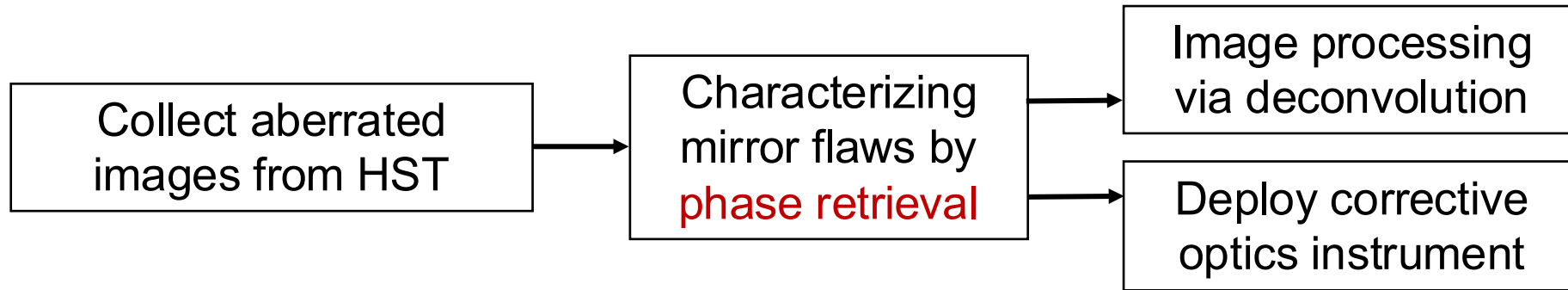
Back to 1990s: HST's Flawed Mirror



Back to 1990s: HST's Flawed Mirror

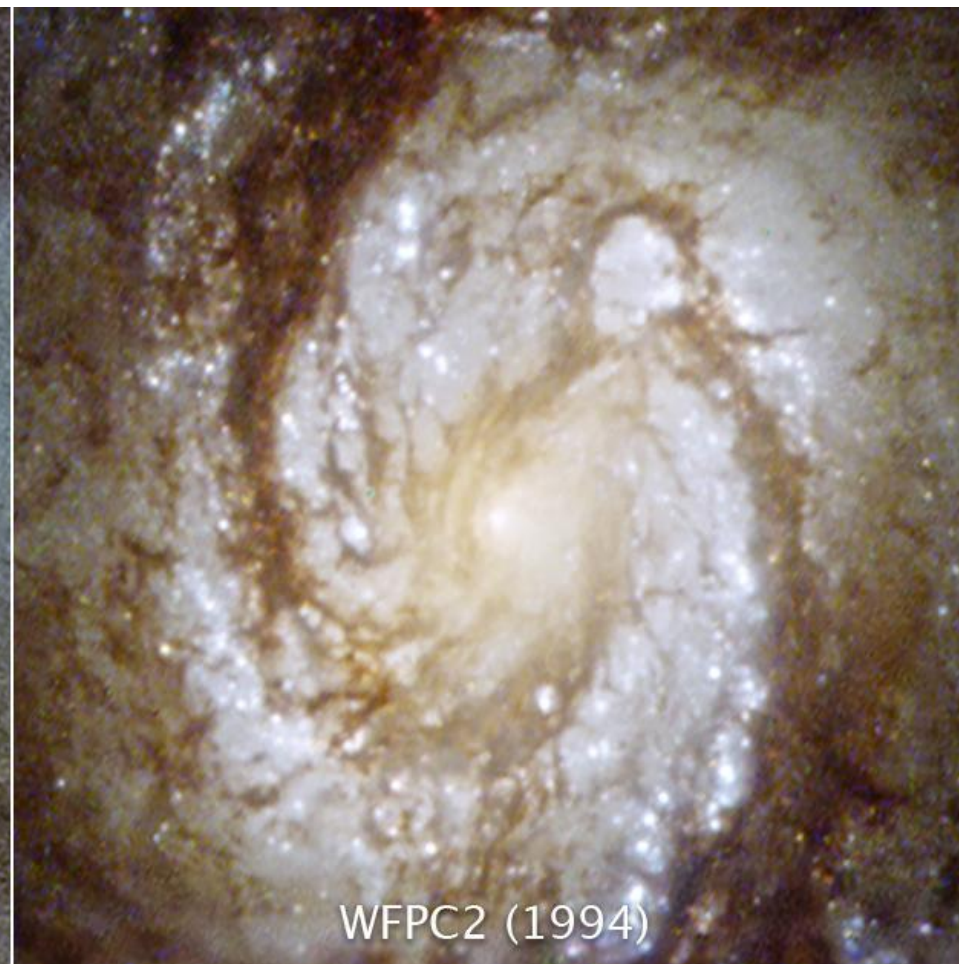
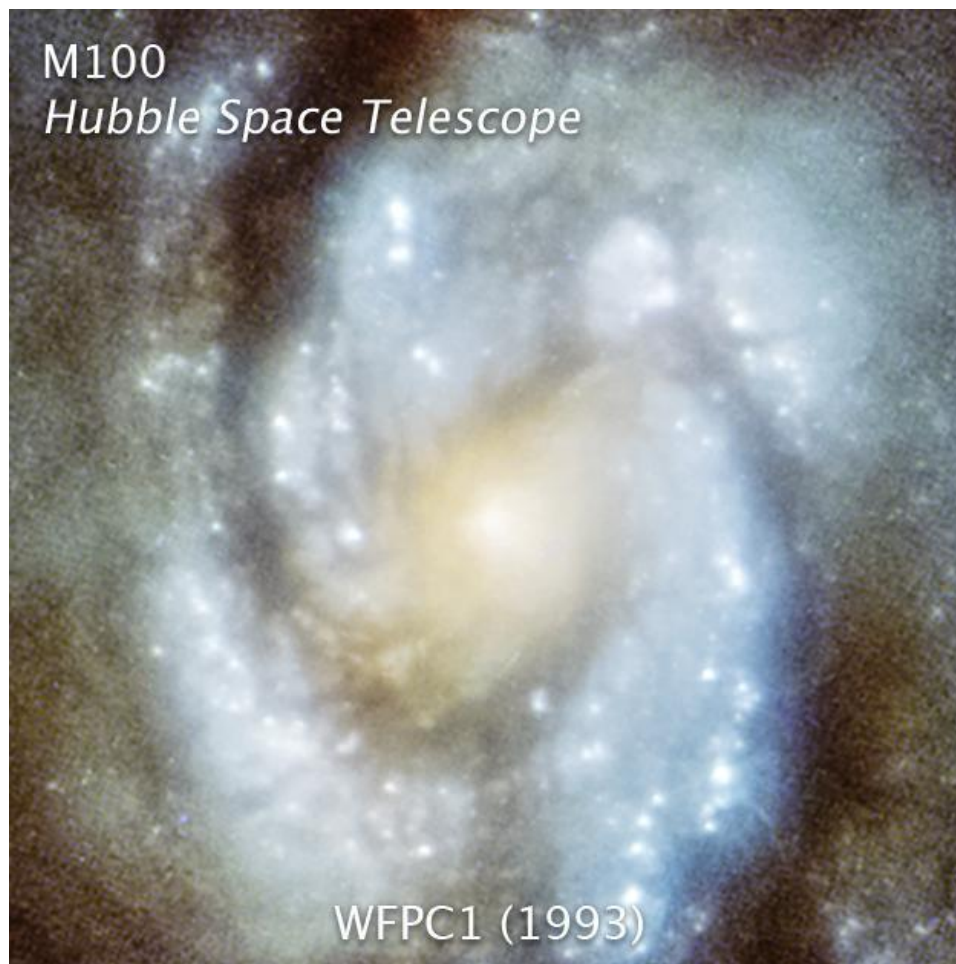


Rescuing Hubble Space Telescope (HST)



1993
HST Servicing
mission

Rescuing Hubble Space Telescope (HST)



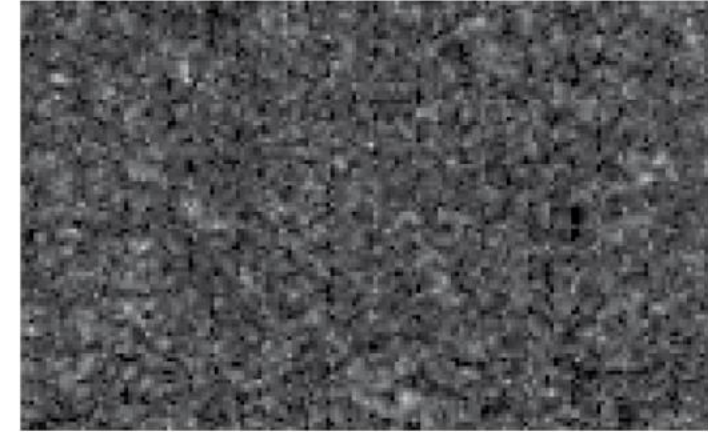
Late 1990s - Vision AO

Aberrations in vision AO

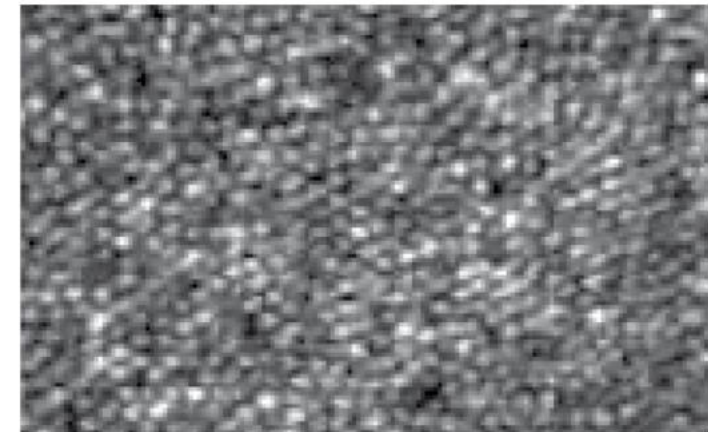
- Eye's imperfect optics
- Diffraction caused by the finite size of the eye's pupil (1–8 mm diameter)
- Diffraction can be minimized by dilating the pupil with mydriatic drops. However, this benefit comes at the cost of additional aberrations exposed by dilation
- AO increases lateral resolution by a factor of up to five
- AO increases sensitivity as it allows a larger pupil to be used and more reflected light to be captured by the ophthalmoscope.

AO-OCT reveals thousands of individual cone photoreceptor cells spaced $4.5\ \mu\text{m}$ apart that would otherwise not be seen

AO off



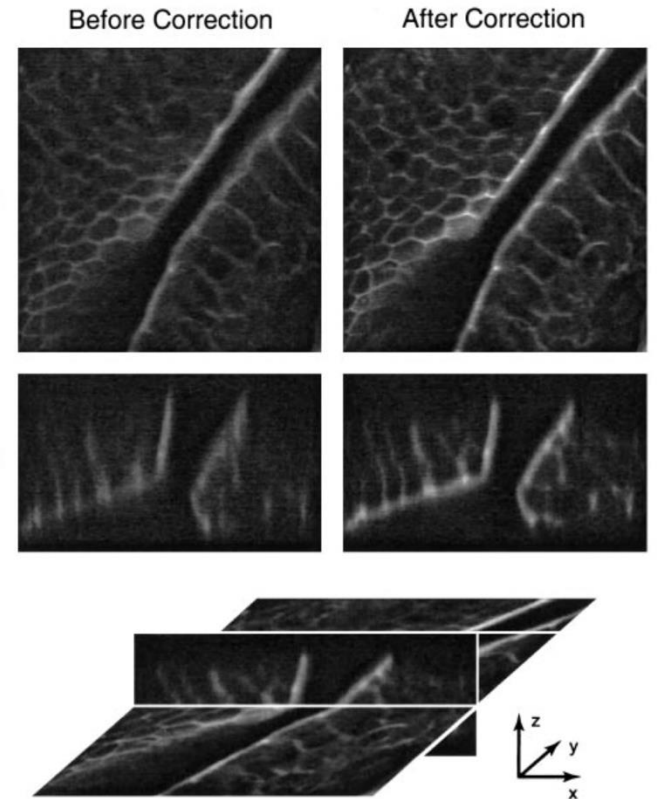
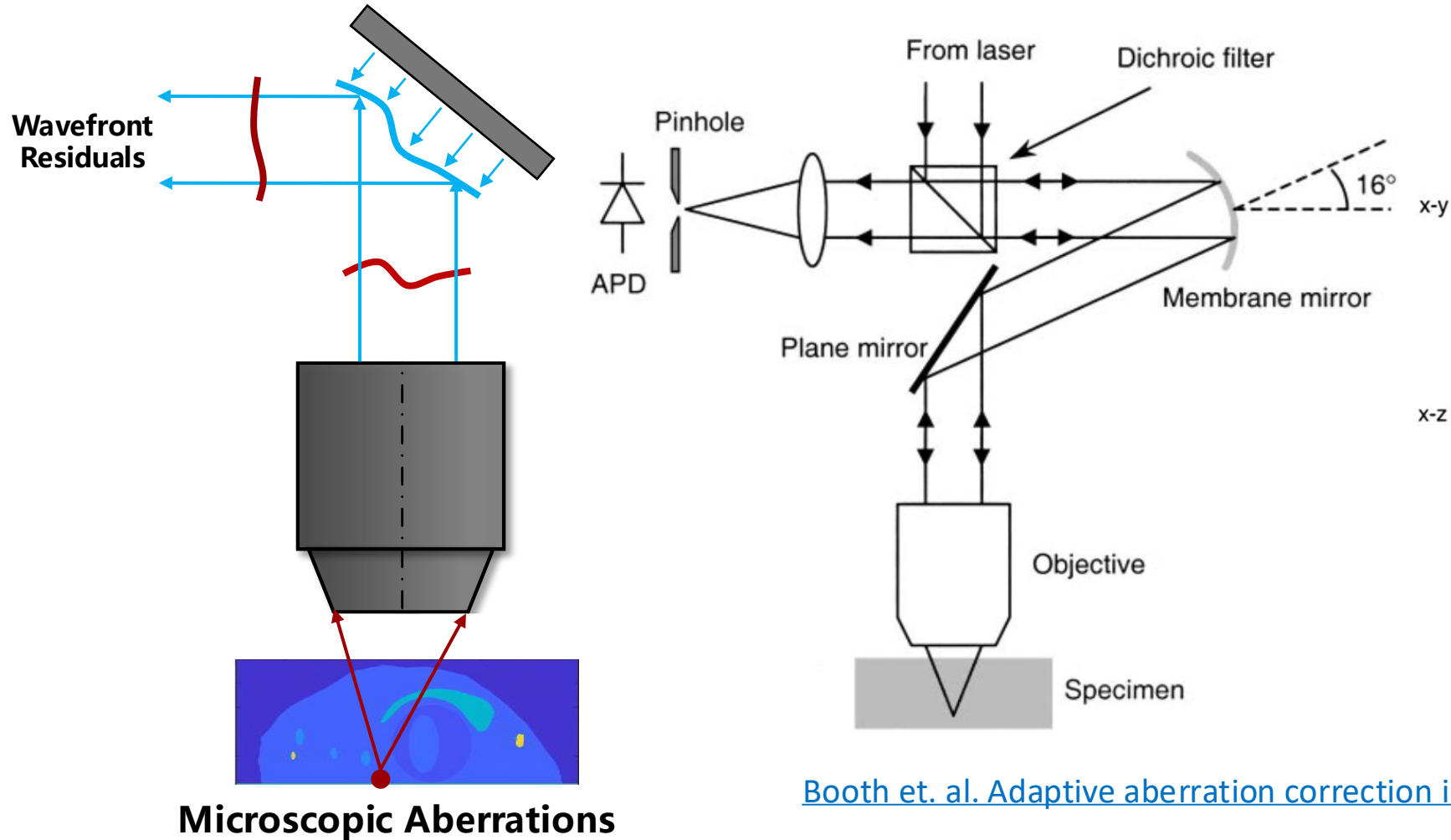
AO on



AO-OCT

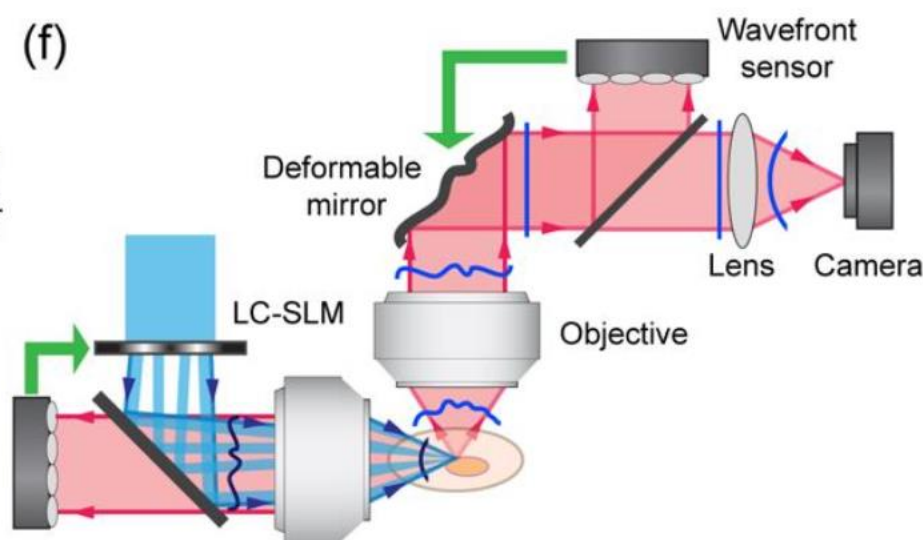
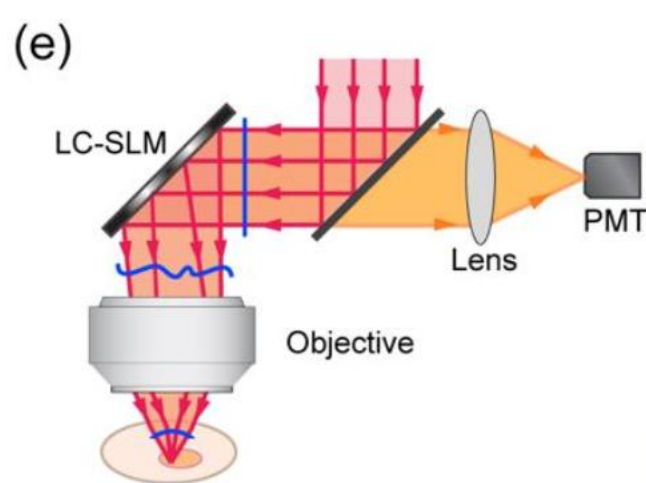
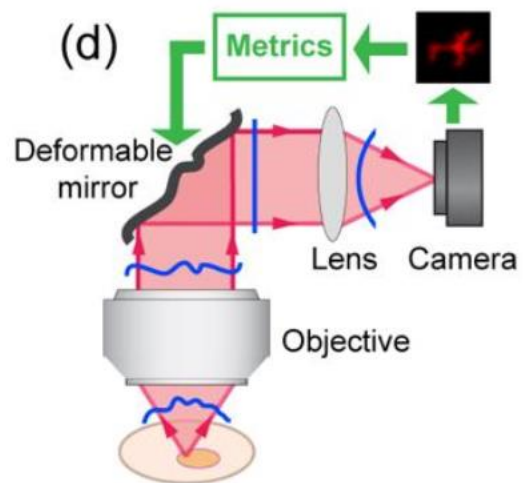
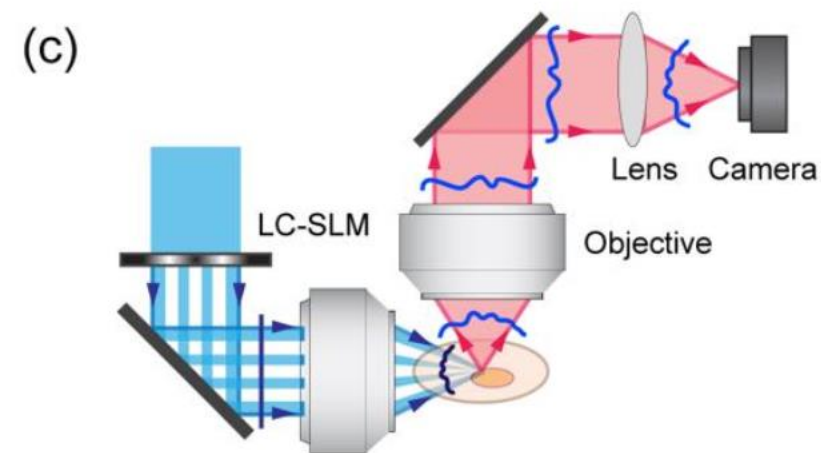
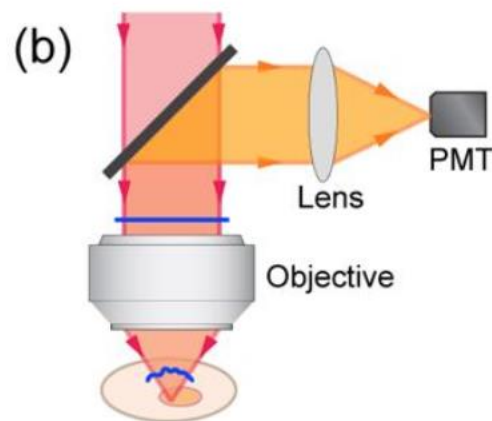
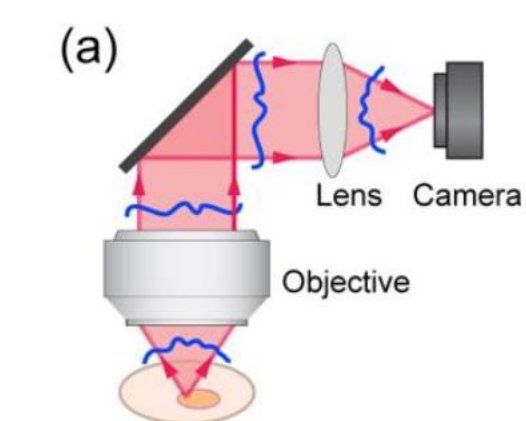
Early 2000s - Microscopic AO

2002 AO was first extended to the field of microscopy.



[Booth et. al. Adaptive aberration correction in a confocal microscope. PNAS \(2002\).](#)

Microscopic AO corrects illumination & emission light

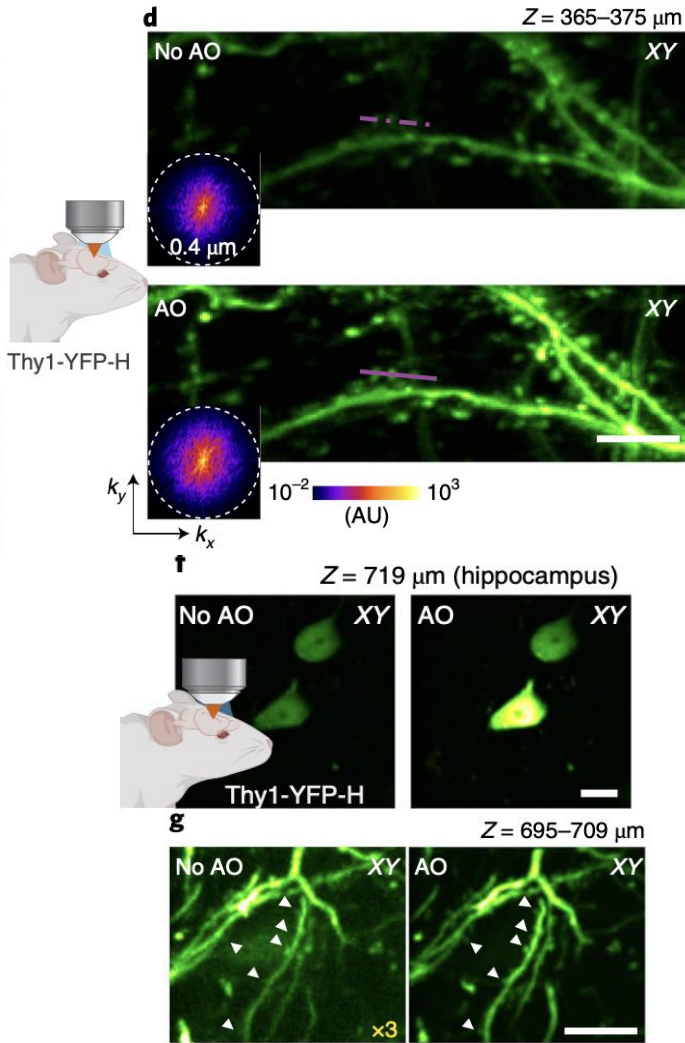


Wide-Field (Emission)

Point Scanning (Excitation)

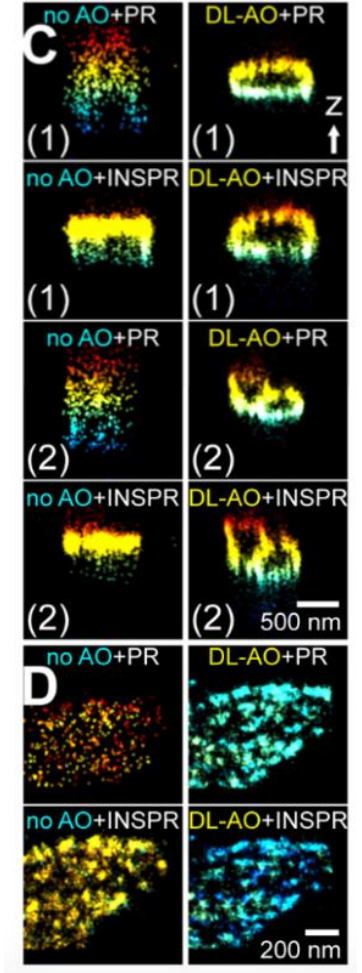
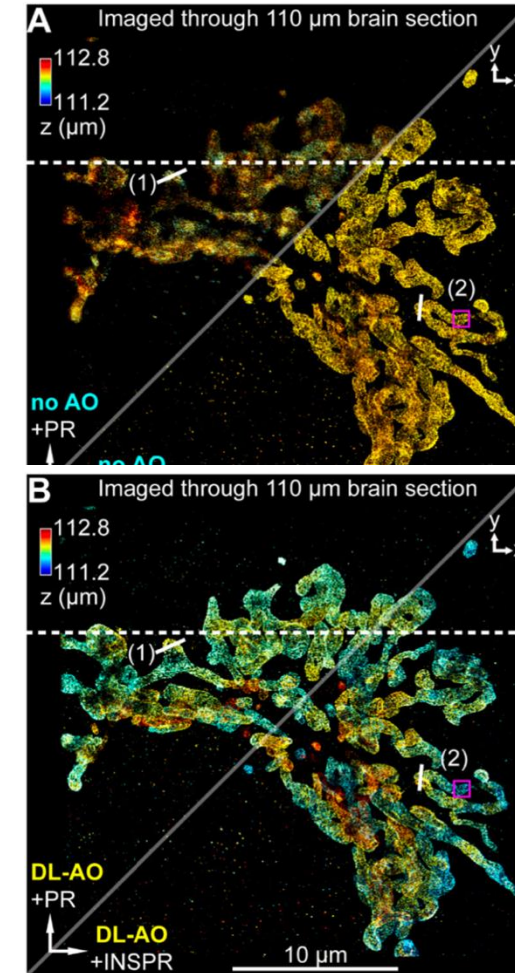
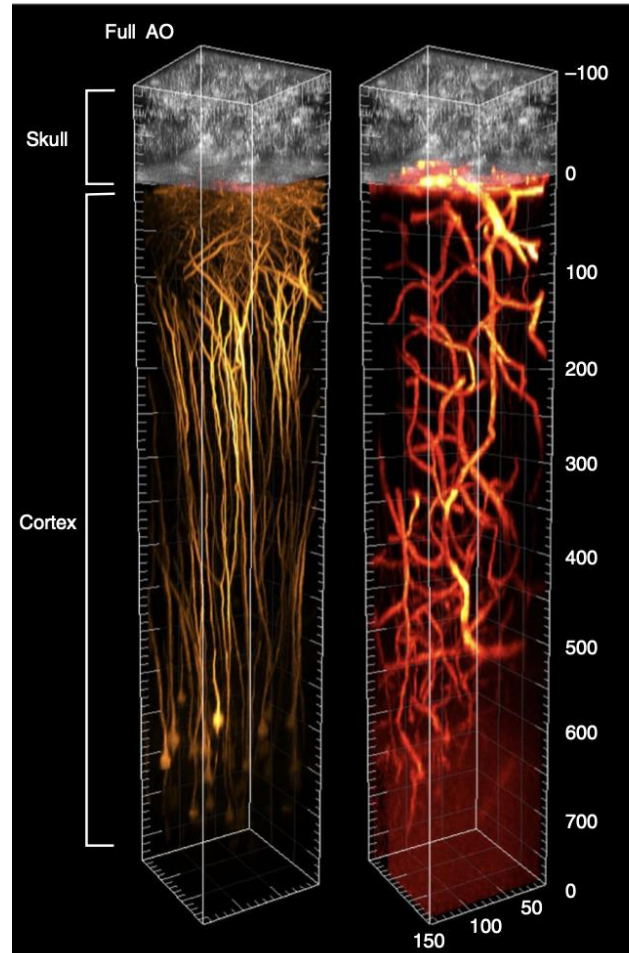
Light-Sheet (Emission & Excitation)

Recent Progress: Multi-photon & Super-resolution



2-photon & 3-photon Microscopy

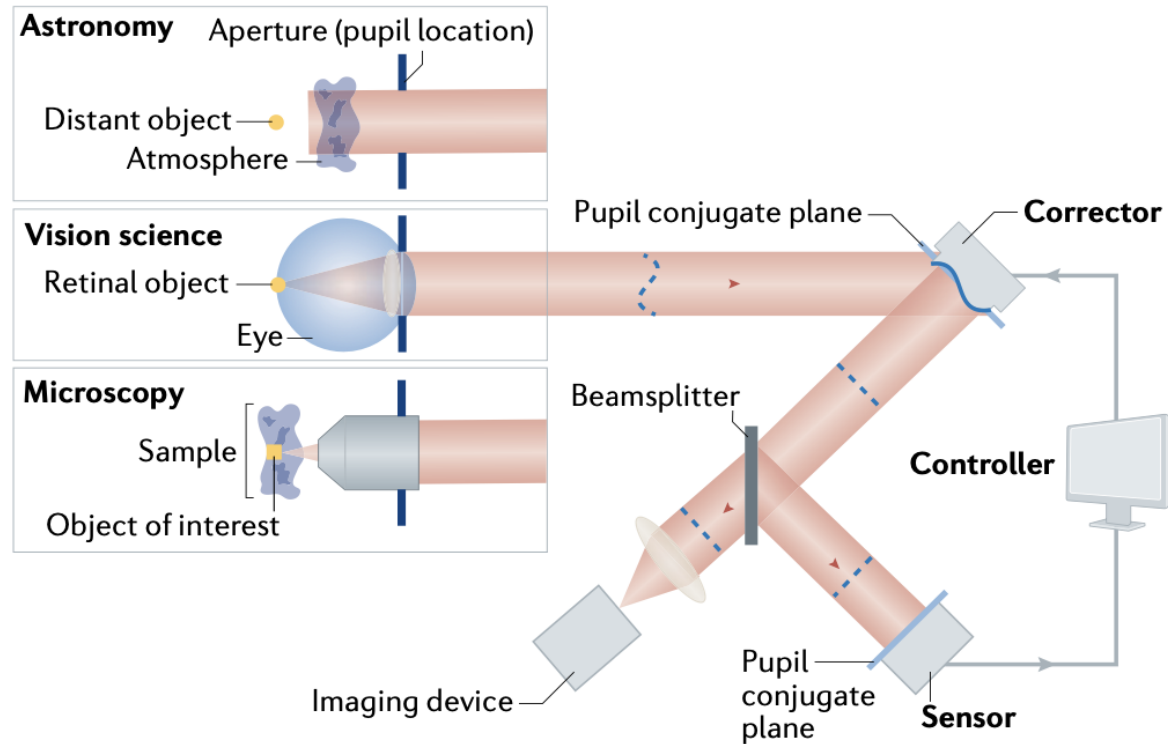
3-photon Microscopy



SMLM Microscopy

Na Ji et al, Nature Methods 2021 Jianan Qu et al, Nature BioTech 2022 Fang Huang et al, Nature Methods 2023

A Representative AO System

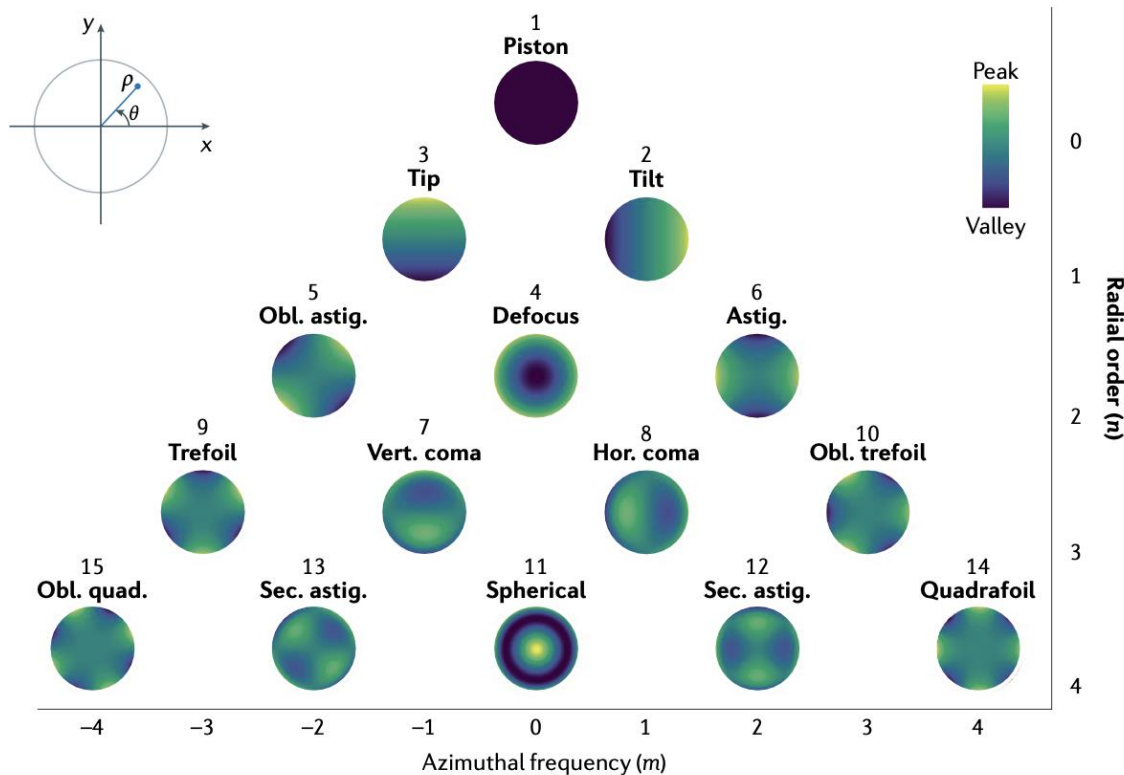


Three main components of AO:

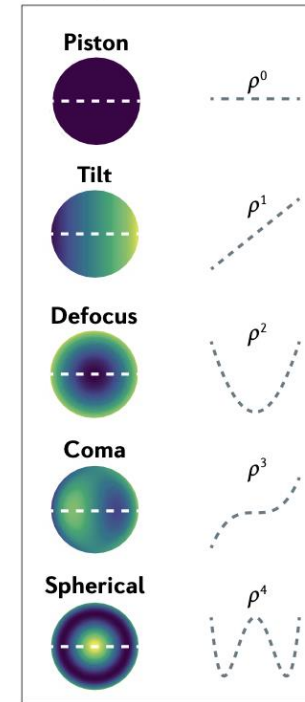
- a sensor to measure the aberrations
- a corrector to compensate the aberrations
- a controller that calculates the required signals sent to the corrector based on the sensor measurements.

The corrector and sensor are typically **conjugate to the pupil plane** of the imaging system, which means that the pupil is imaged on to the corrector and sensor.

Representing the Wavefront - RMS errors & Zernike Polynomials

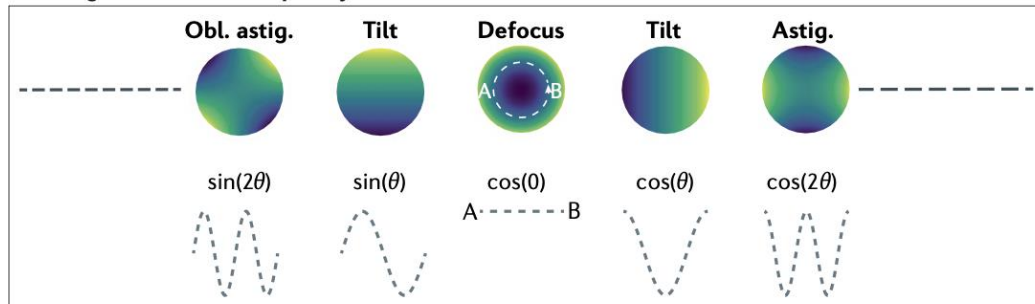


Meaning of radial order



- Aberration modes are often broadly categorized as being of **lower order** or **higher order**.
- The definition of lower order and higher order varies between fields.
- Higher order is typically defined as modes with a **radial order** above three for astronomy, above two for vision science and above four for microscopy.

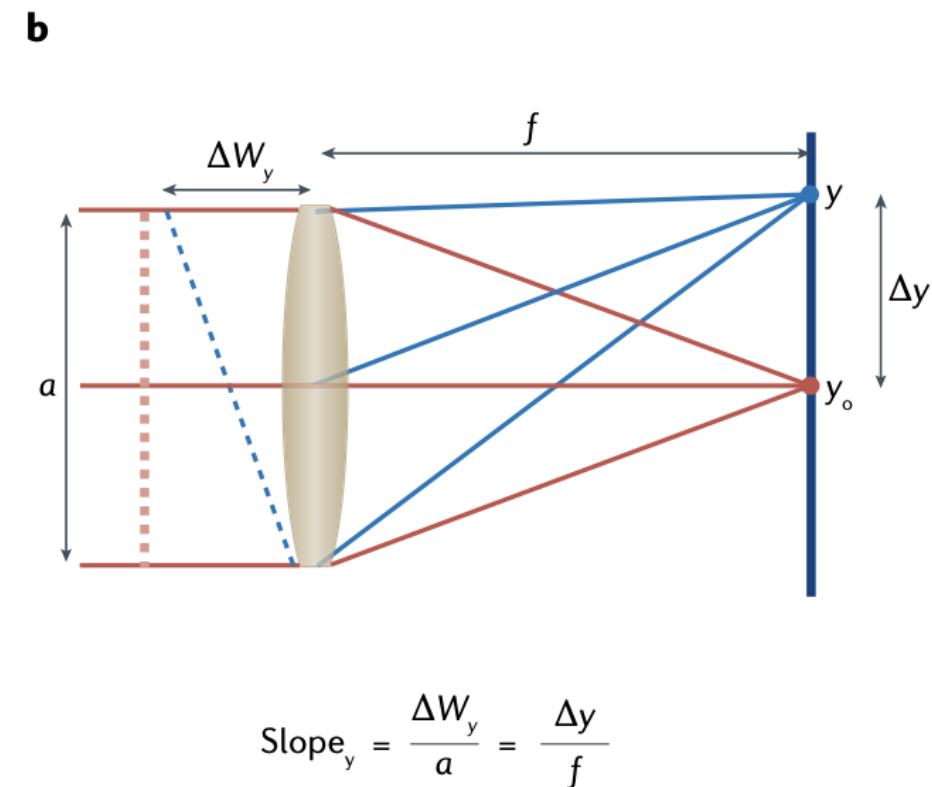
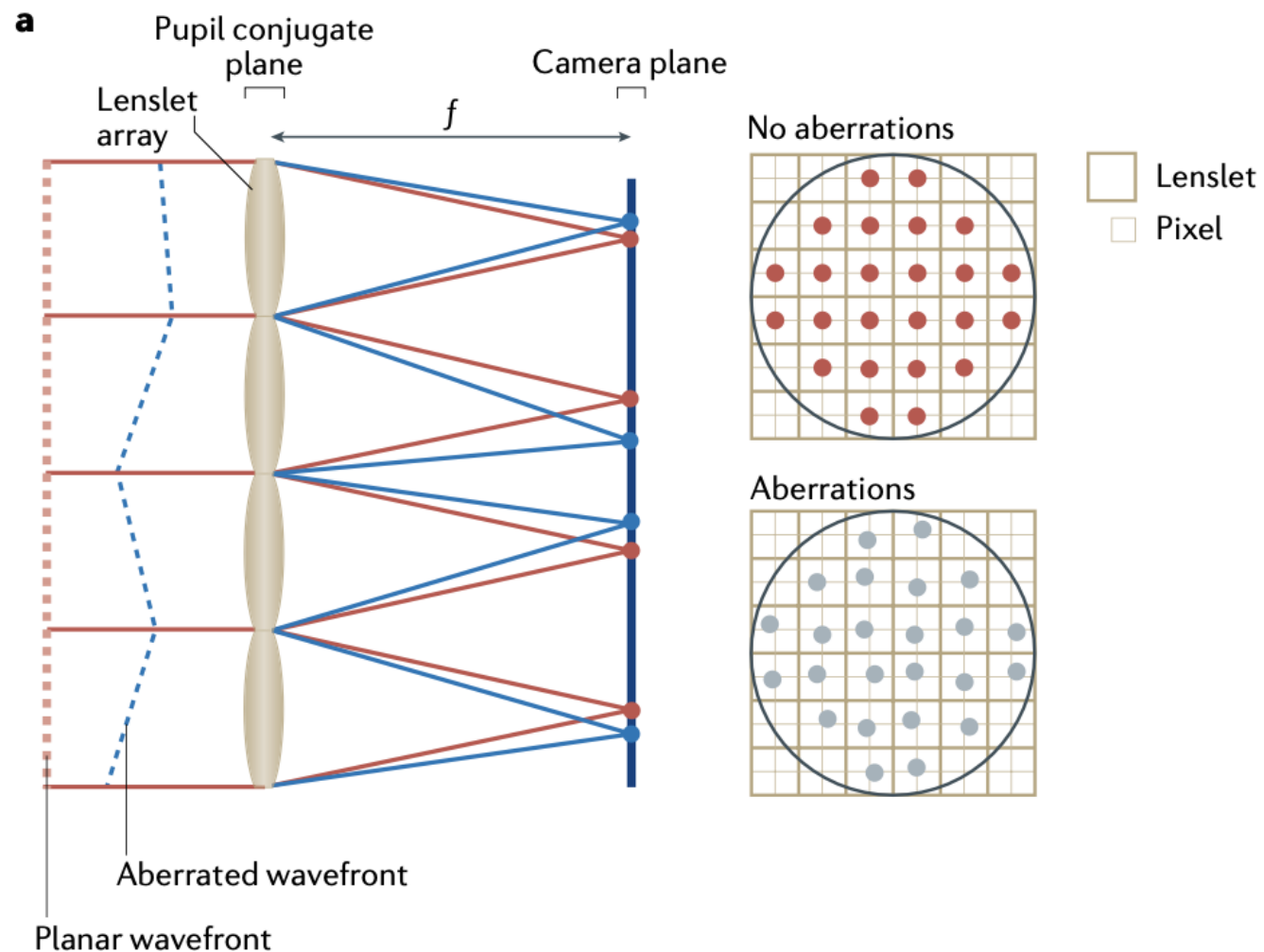
Meaning of azimuthal frequency



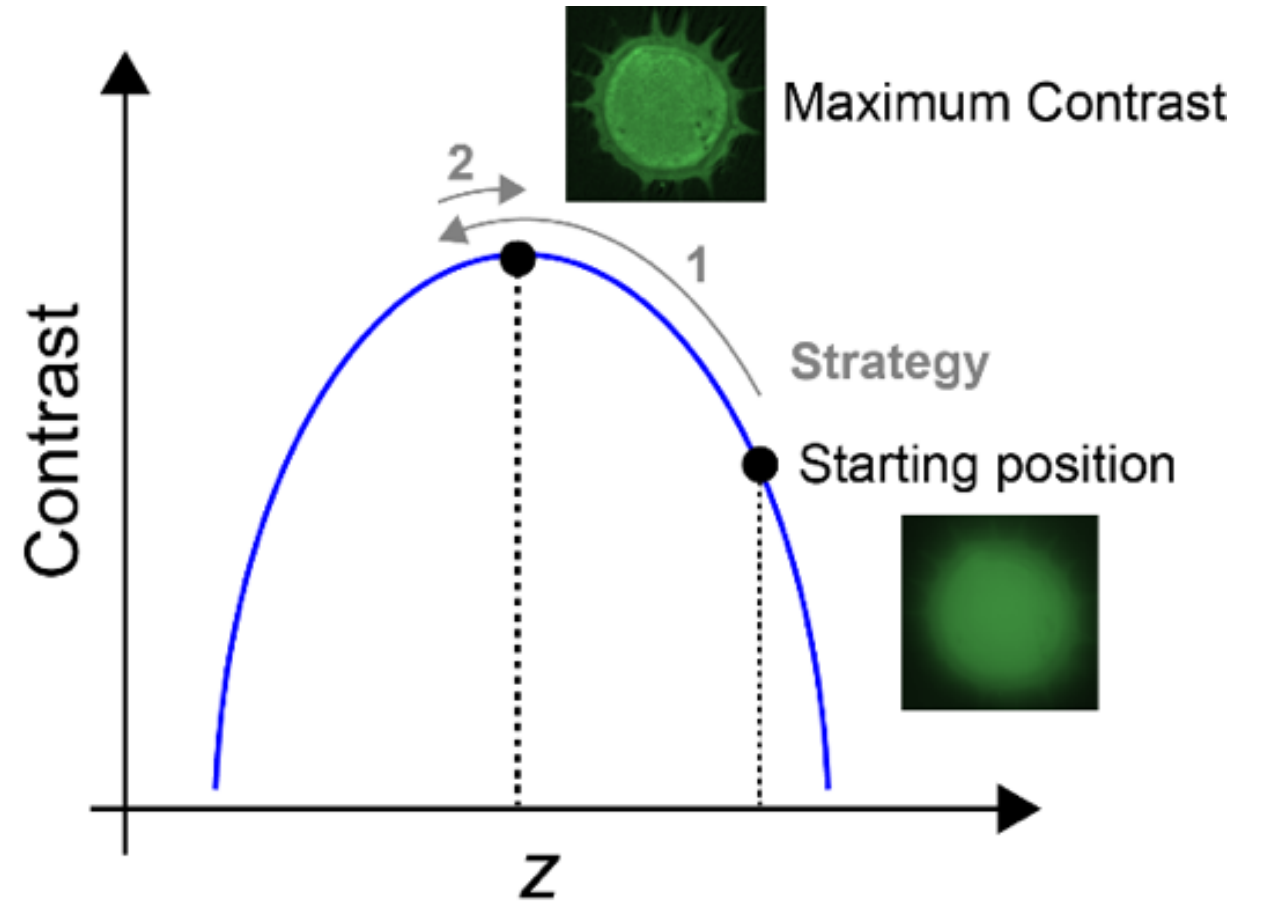
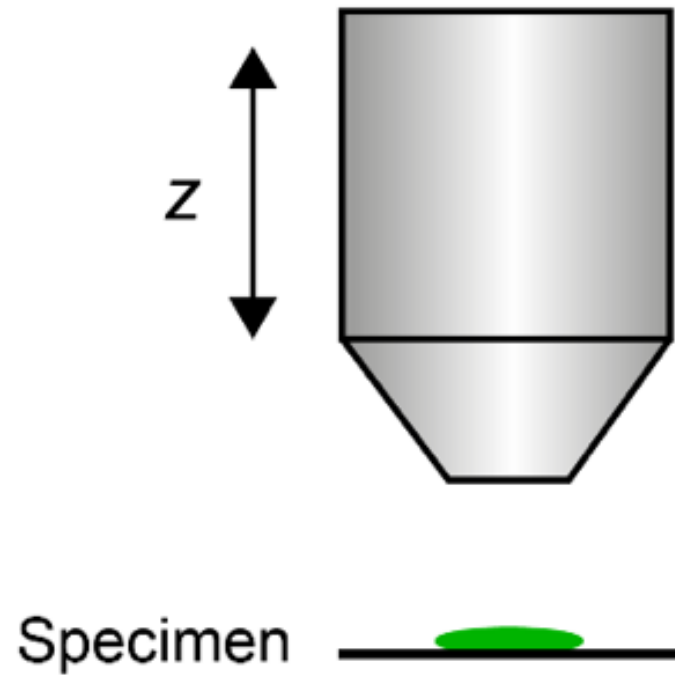
$$Z_n^m(\rho, \theta) = \underbrace{N_n^m}_{\text{Normalization factor}} \times \underbrace{R_n^m(\rho)}_{\text{Radial component}} \times \underbrace{\cos(m\theta)}_{\text{Azimuthal component}}$$

Wavefront Sensing – Direct Sensing

SH sensor



Wavefront Sensing – Indirect Sensing

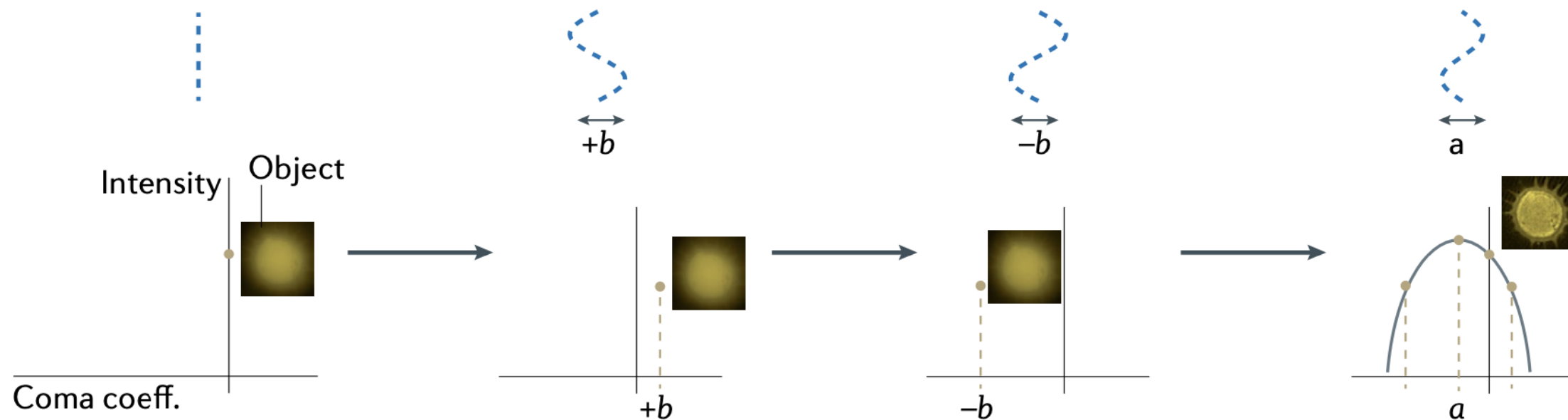


Wavefront Sensing – Indirect Sensing

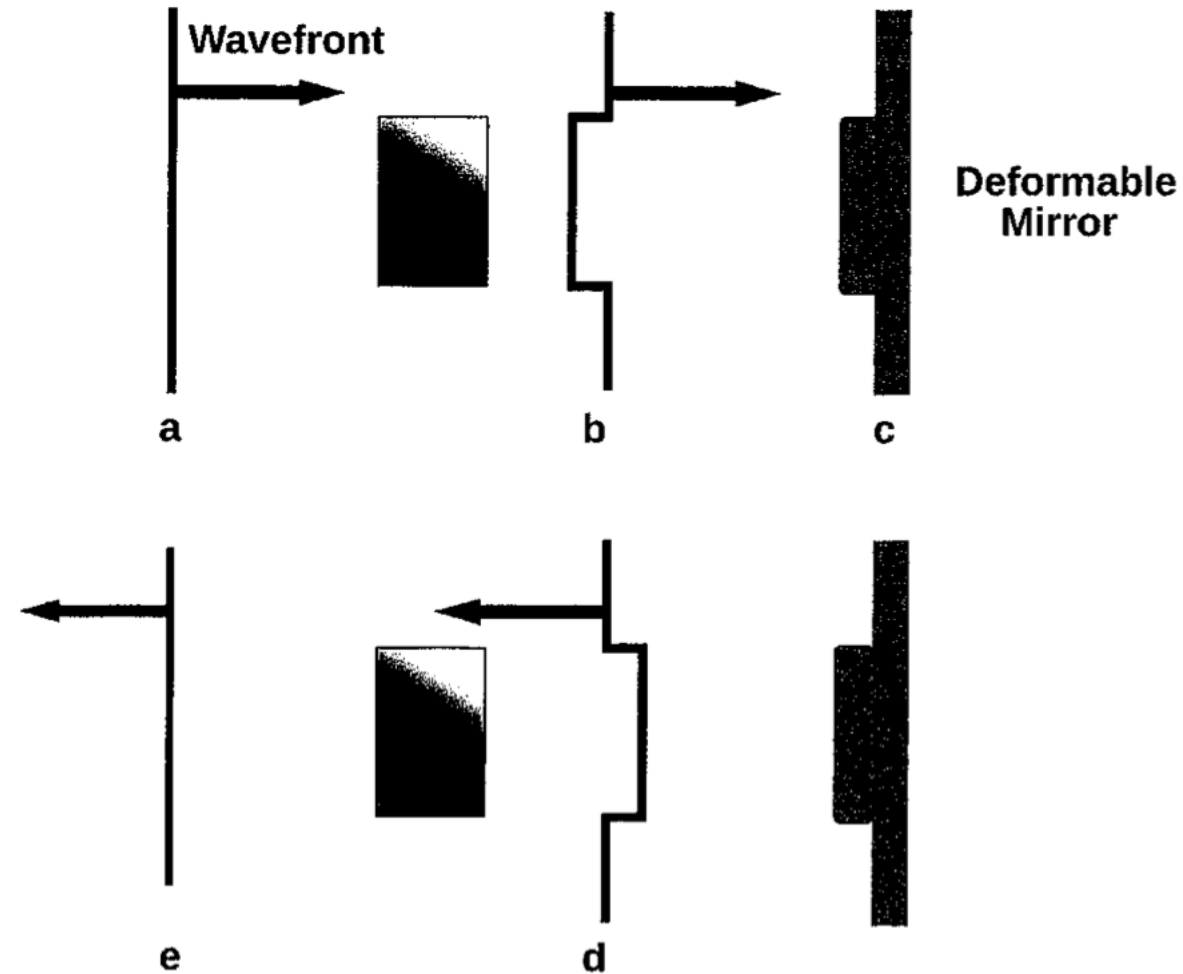
a

Corrector shape

Intensity meas.

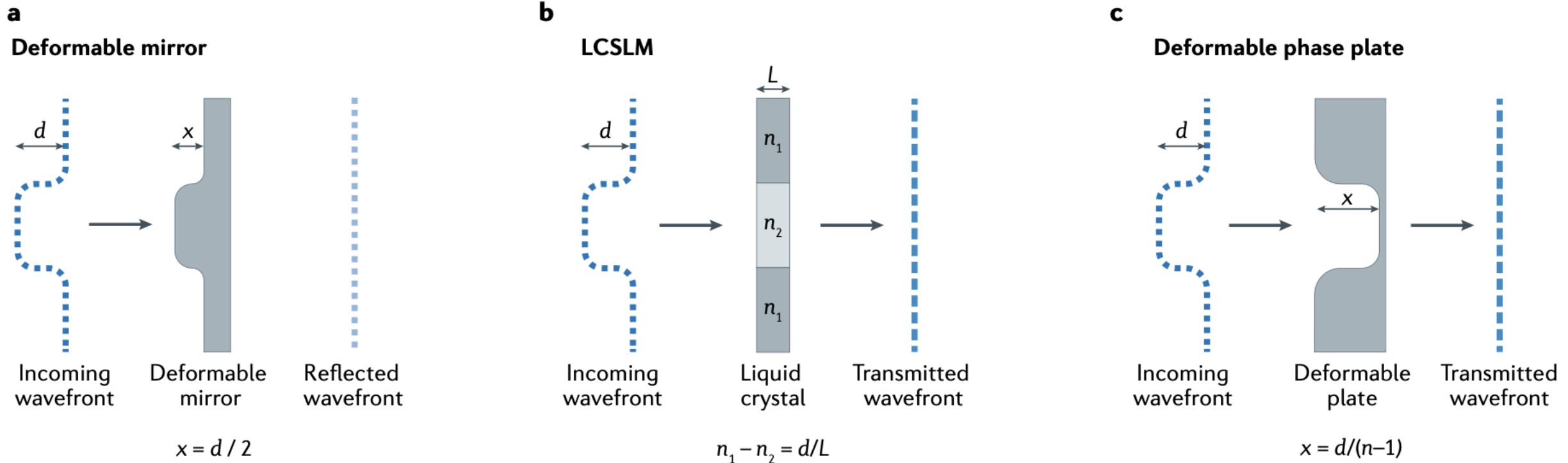


Wavefront Control – Phase Conjugation



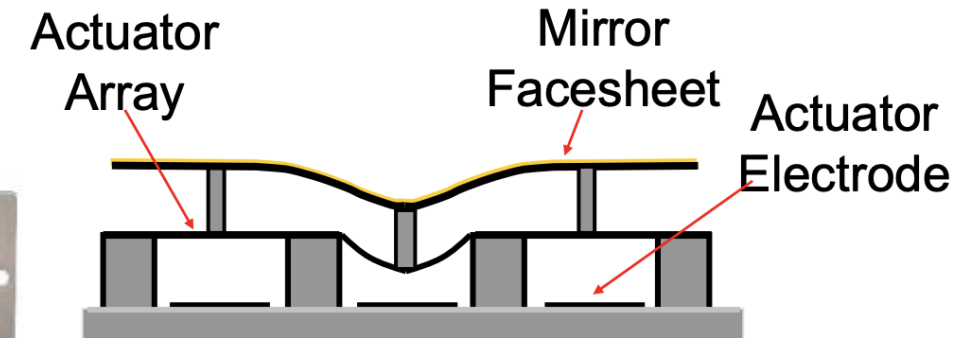
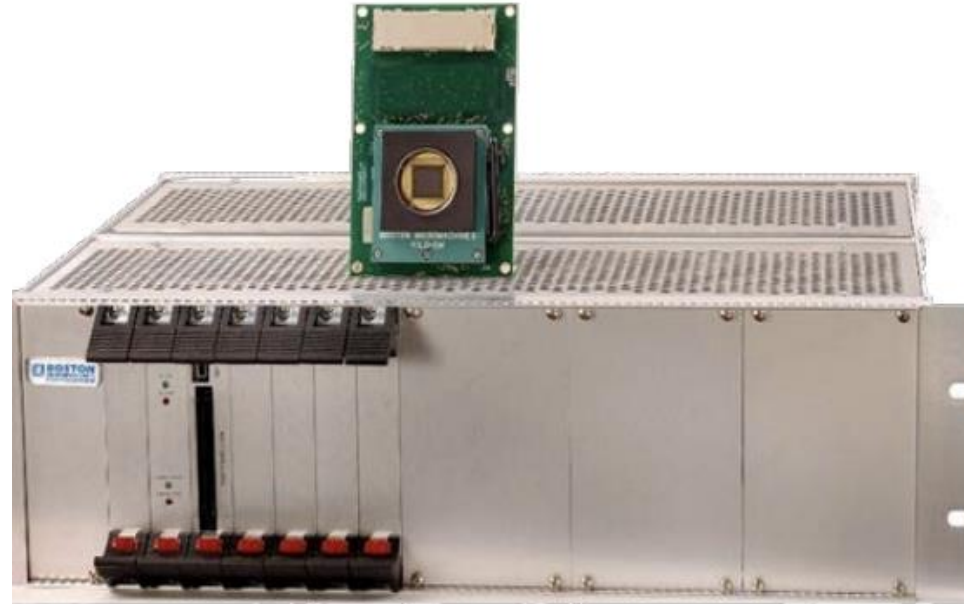
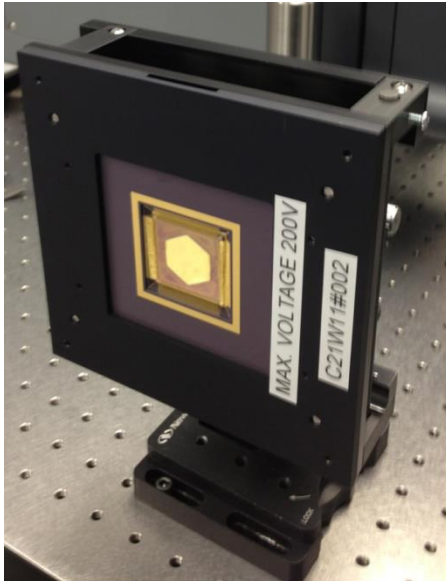
Apply an opposite shape of wavefront error on the corrector

Wavefront Control - Correctors



- a. **Deformable mirrors (DMs)** consist of a reflective surface that may be continuous or segmented.
- b. **Liquid crystal spatial light modulators (LCSLMs)** consist of pixels that are able to change their refractive index, n .
- c. **Deformable phase plates** are fluidic devices that are able to change their shape.

Wavefront Control - Correctors



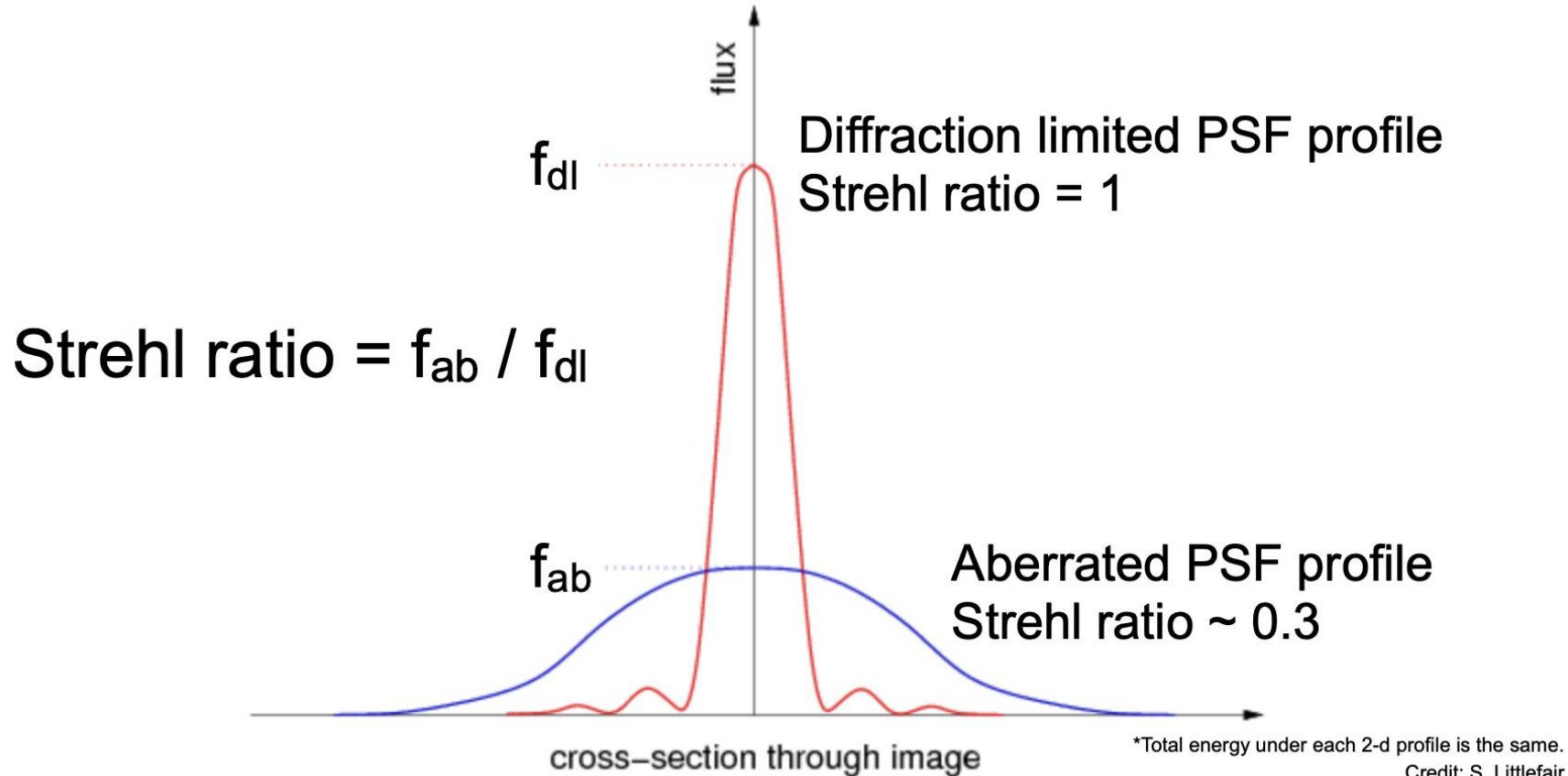
Boston Micromachines Corporation (BMC)
MEMS DMs

- a. **Deformable mirrors (DMs)** consist of a reflective surface that may be continuous or segmented.
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Basic Concept – Strehl Ratio

Strehl Ratio: The ratio of the intensity of the peak of the aberrated point spread function (PSF) to that of the diffraction-limited PSF.

To achieve diffraction-limited resolution, which is equivalent to **a Strehl ratio of 0.8 or above**, AO must reduce this distortion to less than **$\lambda/14$** .



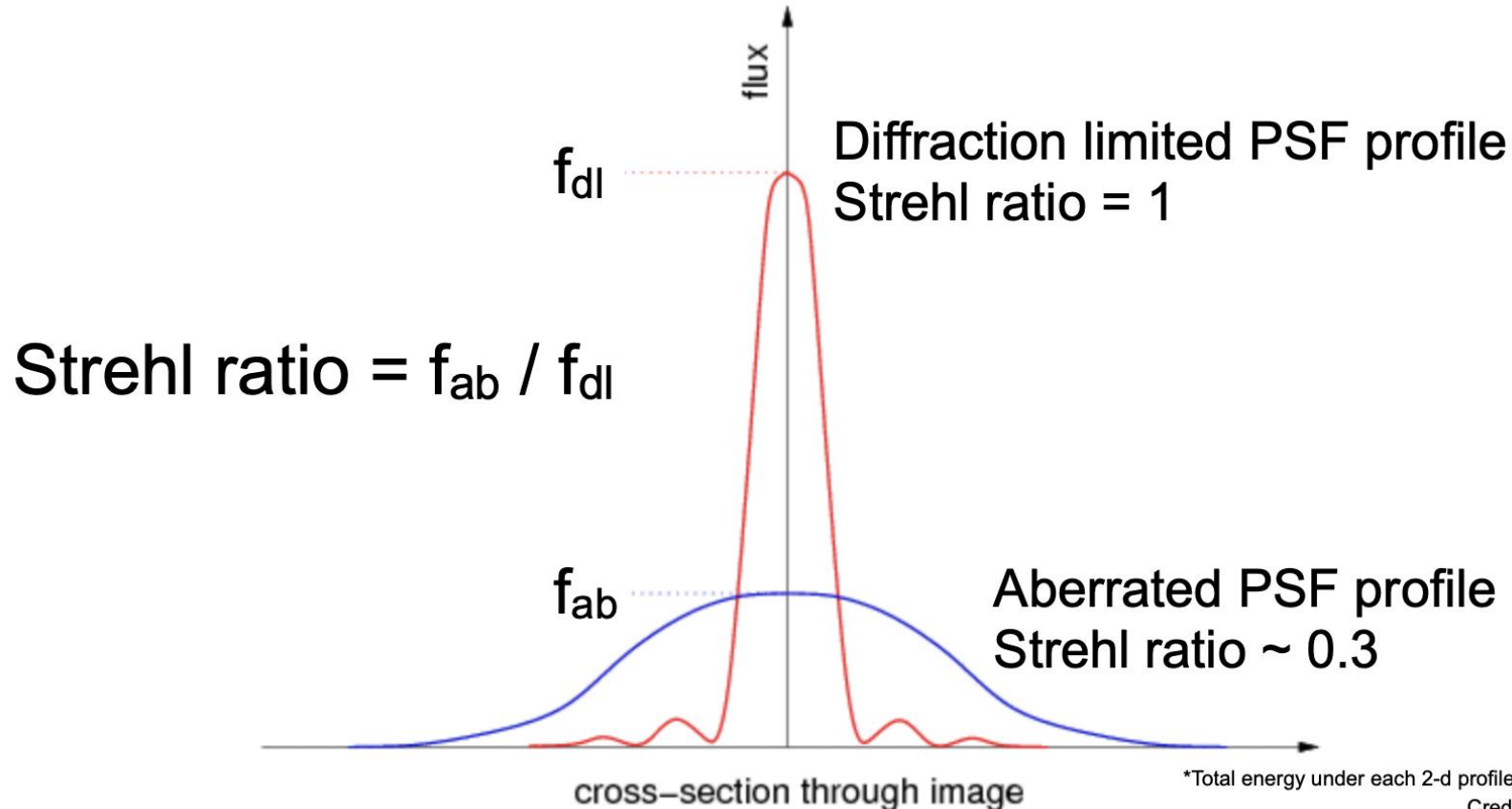
Basic Concept – RMS Error

RMS wavefront error is a useful descriptor to capture the severity of the aberrations in a system.

$$RMS = E[(W - E[W])^2]^{1/2}$$

The Maréchal Approximation: $\text{Strehl Ratio} \sim \exp(-\sigma^2) = \exp(-(2\pi RMS/\lambda)^2)$

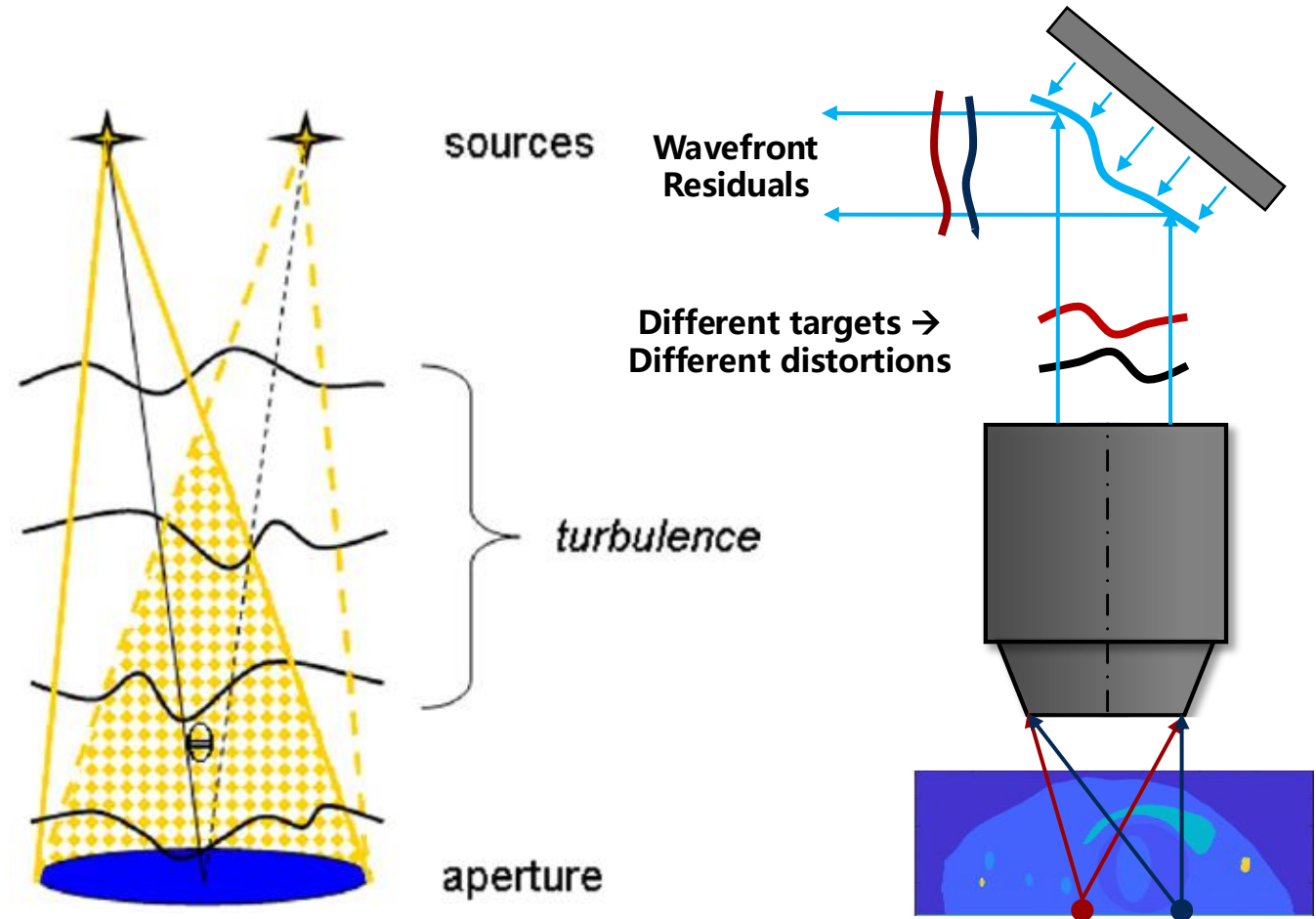
Another empirical rule: $\text{phase variance} = \sigma^2 = 1.03(D/r_0)^{5/3}$



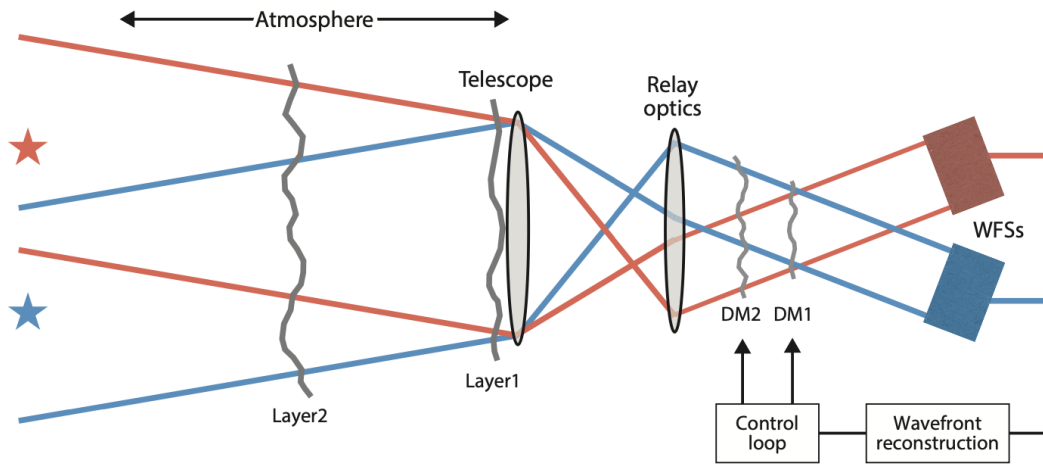
Basic Concept – Isoplanatic Patch

Isoplanatic patch: The area over which the aberrations can be considered invariant.

- **astronomy:** 1.5 to 2 arcsec at $0.5\mu\text{m}$
Isoplanatic path $\propto 1.03(D/r_0)^{6/5}$
- **vision science:** 1° , or $300\mu\text{m}$ at the retina
- **microscopy:** hundreds of microns, in mouse brain for example, to a few microns, such as inside tissue with high curvature or complexity, for example, zebrafish larvae or *Caenorhabditis elegans*.



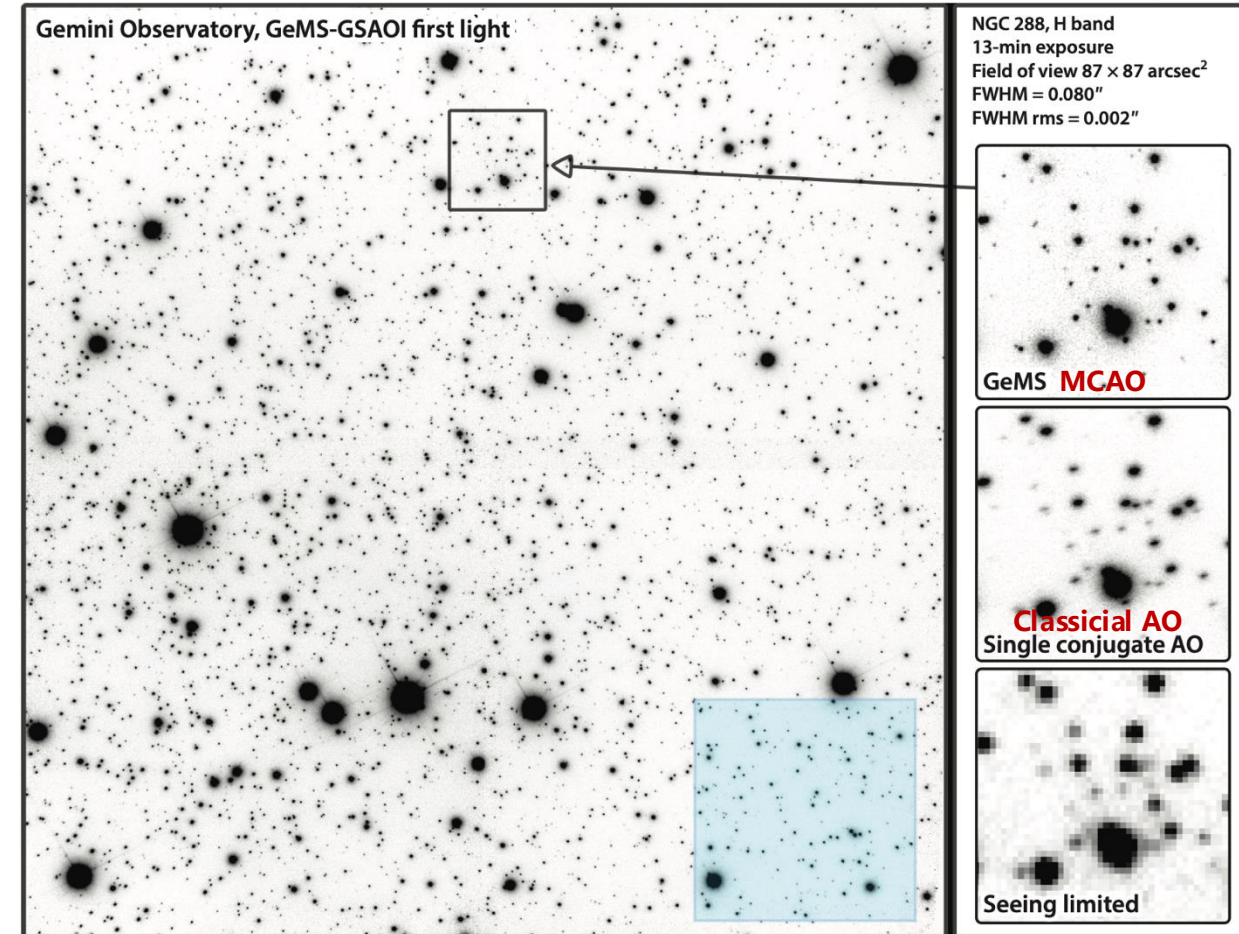
Classification of AO systems: Single-Conjugate v.s. Multi-Conjugate



Single Conjugate AO (SCAO), i.e. classical AO,
→ a single DM conjugated to turbulence in a single atmospheric layer.
→ a moderate FOV at a moderate Strehl ratio.

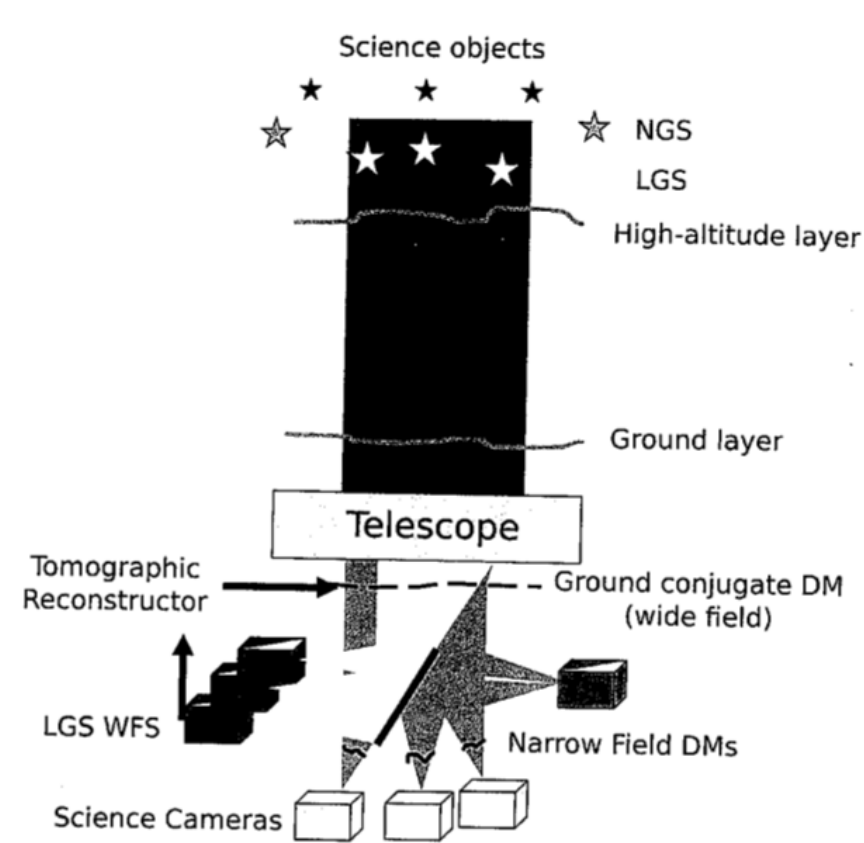
Ground Layer AO (GLAO)
→ a single DM conjugated to ground level.
→ Large FOV at low Strehl ratio

Multi-conjugate AO (MCAO)
→ multiple WFSs and DMs conjugated to multiple atmospheric layers and requires multiple guide stars.
→ Large FOV at moderate Strehl ratio

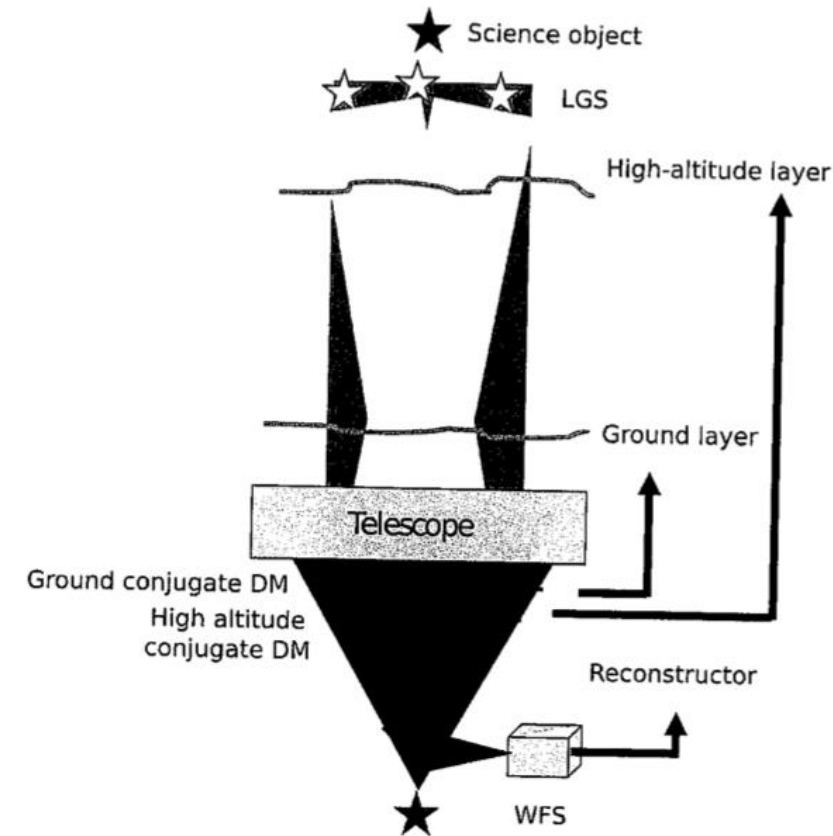


Francois Rigaut et al, MNRAS 2014

Classification of AO systems: Multi-Object v.s. Multi-Conjugate

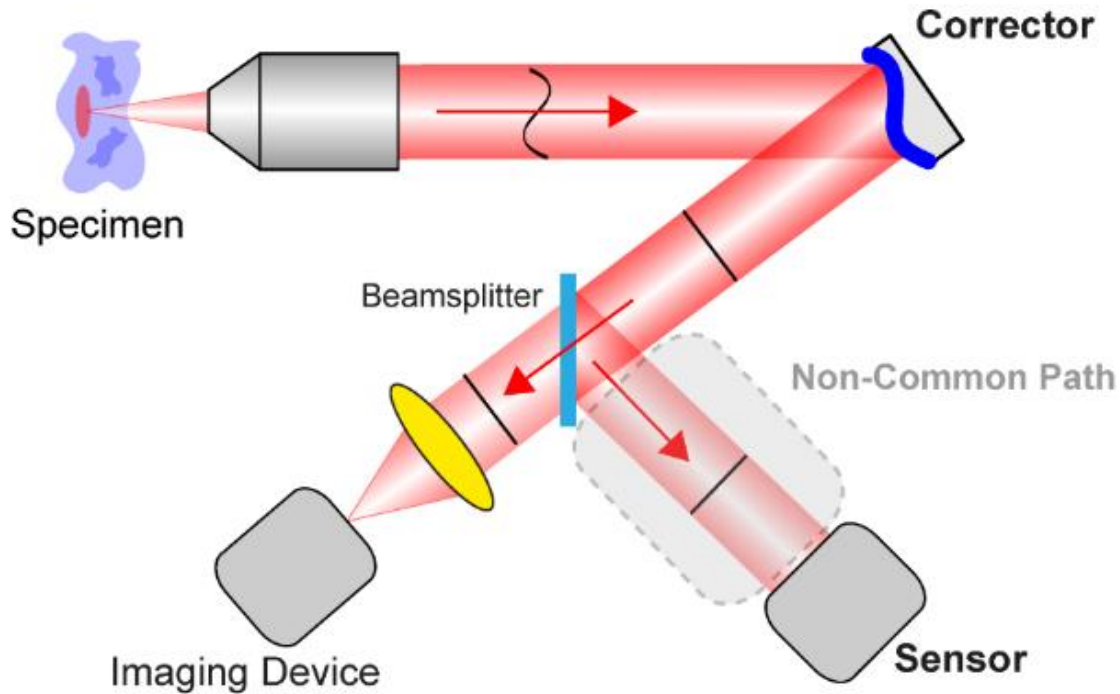


MOAO



MCAO

Classification of AO systems: Sensorless or Sensor-based



Sensor-based AO consisted of a wavefront sensor for aberration measurement.

Sensorless AO requires no sensor, but infers the wavefront aberrations from a sequence of images.

Pros of sensorless AO:

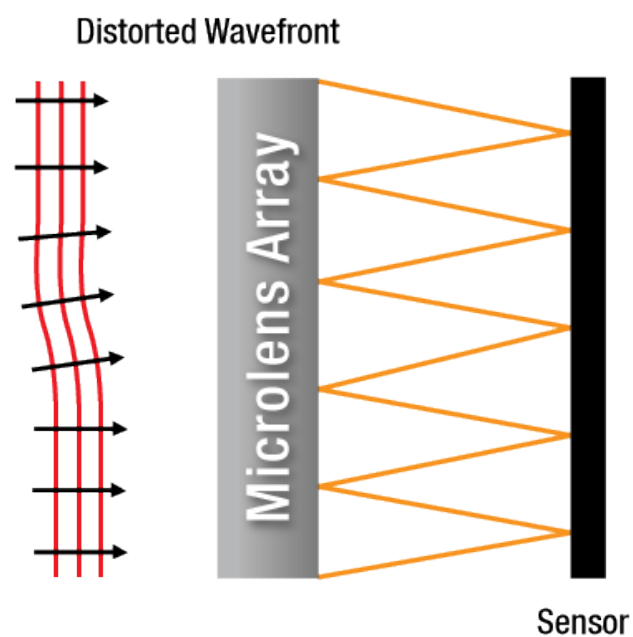
- It requires no extra hardware → low cost
- There are no non-common path errors,
- All light is used for imaging → good for low light case
- It can be used in situations where sensing may not be possible owing to the nature of the light returning from the sample → deep tissue imaging

Cons of sensorless AO:

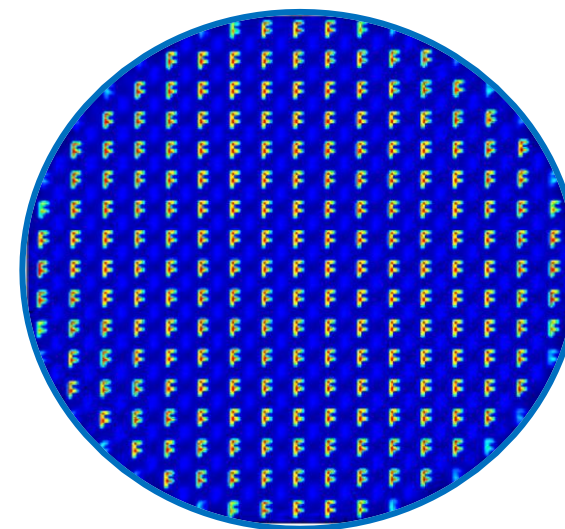
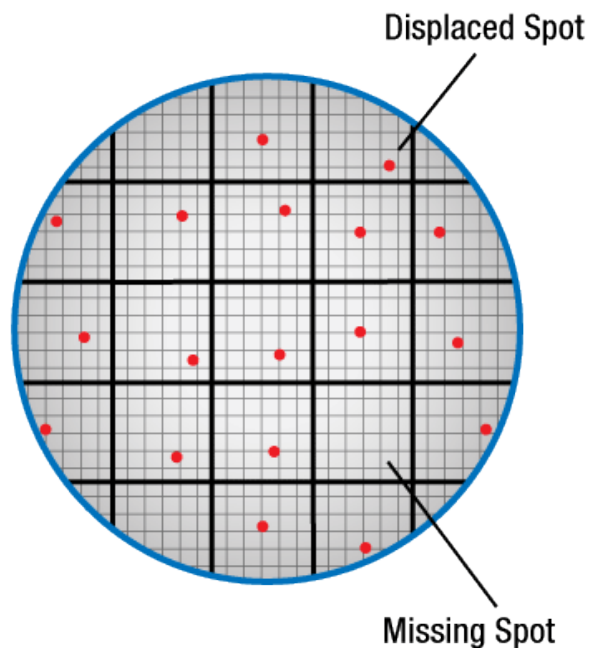
- Speed → photobleaching, live specimen motion, mechanical drift

<https://aomicroscopy.org/sensorless-ao>

Classification of AO systems: Point v.s. Extended Source

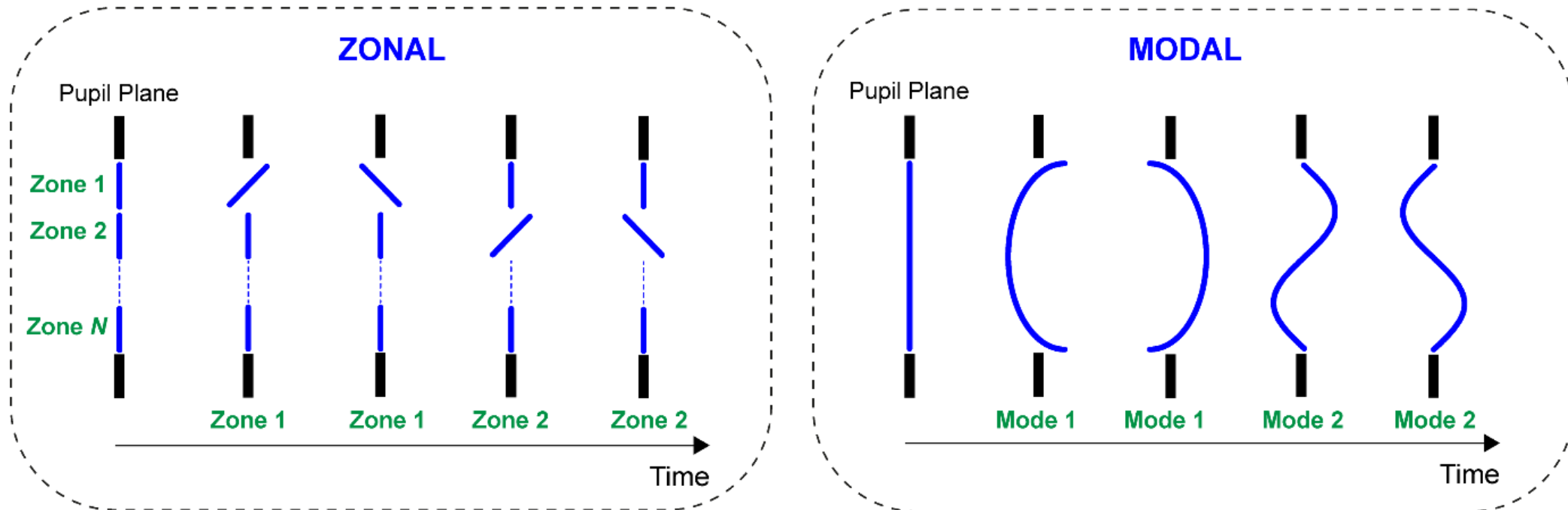


Point Source



Extended Source

Classification of AO systems: Zonal v.s. Modal Correction



The wavefront errors can be corrected either **zonally** or **modally**.

In the **zonal case**, the **pupil** is considered to consist of **segments** that are individually controlled. Zonal techniques are typically carried out using phase-only liquid crystal spatial light modulators (**LC-SLMs**) or **segmented deformable mirrors**. Piston-only or piston, tip and tilt are modulated, depending on the corrector.

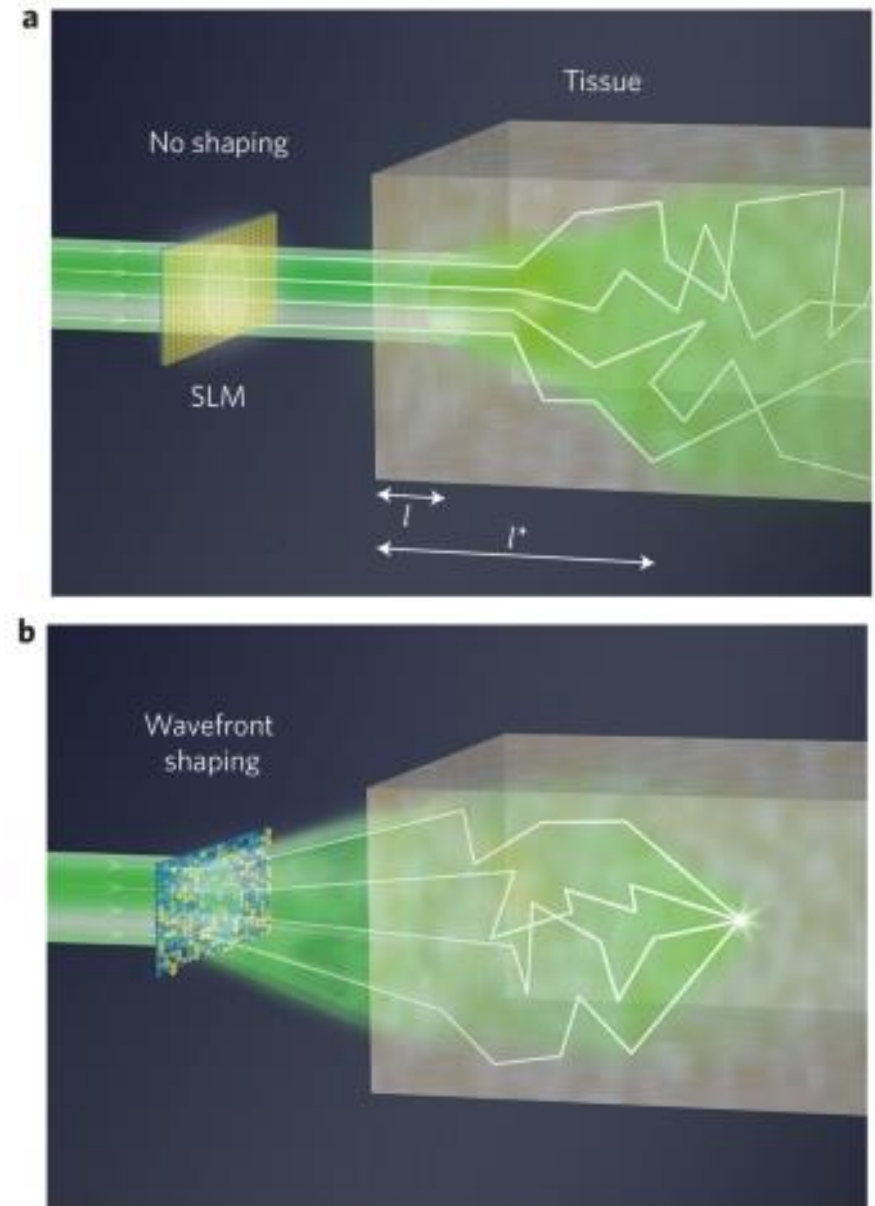
In the **modal case**, the **wavefront shape across the entire pupil** is changed sequentially. Consequently, this method is more suitable for continuous surface correctors such as **continuous surface deformable mirrors**. Each mode could be a Zernike polynomial, for example.

Classification of AO systems: AO v.s. Wavefront Shaping

AO and wavefront shaping are identical, except for the controller's degrees of freedom.

AO is mainly correct for smooth aberrations.

Wavefront shaping compensate both scattering and aberrations (mainly scattering) and have guaranteed convergence even for finding high-dimensional corrections.



Astronomical AO v.s. Vision AO v.s. Microscopy AO

Field	Aberrations	Sensors	Correctors	Control
Astronomy	Main source is wind-driven motion of the atmosphere Wavefront variance decays with Zernike radial order following a power law with exponent $-11/3$ Temporal PSD varies with $f^{-17/3}$ Peak-to-valley wavefront is $1\text{--}3\text{ }\mu\text{m}$ rms Isoplanatic patch size is $1.5\text{--}2$ arcsec at 500 nm Correcting between 15 and 70 Zernike radial orders is typical	SH sensor most widely used SH lenslets to actuators ratio $\sim 1:1$ Pyramid sensors increasingly used	Continuous surface DMs most widely used Typical number of DM actuators: $100\text{s--}1,000\text{s}$	Closed-loop systems by far the most common Closed-loop bandwidth of 50 Hz is typical Integral controller is most widely used
Vision science	Main source is optical imperfections in the crystalline lens and cornea Wavefront variance decays exponentially with Zernike radial order in the range -1.1 to -1.8 Temporal PSD varies with $f^{-1.3} - f^{-1.5}$ Peak-to-valley wavefront is $7\text{--}11\text{ }\mu\text{m}$ for a dilated pupil with no refractive error Isoplanatic patch size is about $300\text{ }\mu\text{m}$ Correcting 10 Zernike radial orders is typical for a dilated pupil	SH sensor most widely used SH lenslets to actuators ratio $3:1\text{--}6:1$	Continuous surface DMs most widely used Typical number of DM actuators <100	Closed-loop systems by far the most common Closed-loop bandwidth of $<2\text{ Hz}$ is typical Integral controller is most widely used
Microscopy	Main source is shape and refractive index inhomogeneity of cells and tissue Aberration magnitudes vary with sample Aberrations are mostly temporally static Peak-to-valley wavefront varies from submicrometre to several microns Isoplanatic patch size is several to 100s of microns Correcting between 7 and 11 Zernike radial orders is typical	Indirect sensing and SH are both common SH lenslets to actuators ratio $\sim 1:1$ Indirect sensing employs both modal and zonal aberration representation	Continuous DMs most widely used LCSLMs used in laser illumination paths Segmented DMs used for high-order scattering compensation Typical number of DM actuators <100 Typical number of SLM pixels 512×512	Open- and closed-loop systems are both common As aberrations are mostly static, closed-loop dynamics are determined by imaging speed

Analysis of AO System

Problem:

Science Target:

Wavelength Range:

Statistics of Aberrations:

...

Hardware&Software:

Sensor:

Corrector:

Algorithm:

Result:

FOV:

Speed:

Resolution:

Contrast: